A Linear-Time Algorithm for the Terminal Path Cover Problem in Cographs

Ruo-Wei Hung
Department of Information Management
Nan-Kai Institute of Technology, Tsao-Tun, Nantou 542, Taiwan
rwhung@nkc.edu.tw

Abstract

Let $G = (V, E)$ be a graph with vertex set V and edge set E and let $\mathcal{T}$ be a subset of V. A terminal path cover $\mathcal{PC}$ of G with respect to $\mathcal{T}$ is a set of pairwise vertex-disjoint paths of G, such that all vertices of G are visited by exactly one path of $\mathcal{PC}$ and all vertices in $\mathcal{T}$ are end vertices of paths in $\mathcal{PC}$. The terminal path cover problem is to find a terminal path cover of G of minimum cardinality with respect to $\mathcal{T}$. The path cover problem is a special case of the terminal path cover problem with $\mathcal{T} = \emptyset$. A graph is a cograph if it can be described by expressions involving operations of merging disjoint graphs and taking complements of graphs. Cographs have arisen in many different areas of applied mathematics and computer science and have been independently rediscovered by various researchers. In this paper, we show that the terminal path cover problem on cographs can be solved in linear time.

1 Introduction

All graphs considered in this paper are finite and undirected, without loops or multiple edges. Let $G = (V, E)$ denote a graph with vertex set V and edge set E. For a graph G, $V(G)$ and $E(G)$ denote the vertex set and edge set of G, respectively. A *path cover* of a graph G is a collection of vertex-disjoint paths $P_1 = (V_1, E_1), \cdots, P_k = (V_k, E_k)$ in G whose union is $V(G)$, where V_i and E_i, for $1 \leqslant i \leqslant k$, are the vertex and edge sets of path P_i, respectively, i.e., $V_i \cap V_j = \emptyset$ for $i \neq j$ and $\cup_{i=1}^{k} V_i = V(G)$. The *path cover problem* is to find a path cover of a graph of minimum cardinality. It is evident that the path cover problem for general graphs is NP-complete since finding a path cover, consisting of a single path, corresponds directly to the Hamiltonian path problem [15]. The Hamiltonian path problem on some special classes of graphs, including bipartite graphs [20], split graphs [16], chordal bipartite graphs [26], strong chordal split graphs [26], undirected path graphs [3], and directed path graphs [28], has been shown to be NP-complete. Hence the path cover problem on these above graphs and their superclasses of graphs is also NP-complete. However, it admits polynomial time algorithms when the input is restricted to be in some special classes of graphs, including trees [24], block graphs [25], interval graphs [2, 8], circular-arc graphs [17], cographs [9, 10, 18, 23], bipartite distance-hereditary graphs [36], bipartite permutation graphs [34], and cocomparability graphs [12]. The path cover problem has found some applications in database designing, networking, establishing ring protocol, VLSI designing, code optimization [6], and mapping parallel programs into parallel architectures [24, 31].

Let P be a path of a graph G and let $\mathcal{T}$ be a subset of $V(G)$. The first and last vertices visited by P are called the *path-start* and *path-end* of P, denoted by *start*(P) and *end*(P), respectively. Both of them are *end vertices* of P. A *terminal path cover* of a graph G with respect to $\mathcal{T}$ is a path cover of G such that all vertices in $\mathcal{T}$ are end vertices of paths in the path cover. A minimum terminal path cover of G with respect to $\mathcal{T}$ is a terminal path cover of G of minimum cardinality. Denote by $\pi(G, \mathcal{T})$ the cardinality of a minimum terminal path cover of G with respect to $\mathcal{T}$. For simplicity, we will use $\pi(G)$ to denote $\pi(G, \mathcal{T})$ if $\mathcal{T} = \emptyset$. Given a graph G and a subset $\mathcal{T}$ of $V(G)$, the *terminal path cover problem* is to find a minimum terminal path cover of G of size $\pi(G, \mathcal{T})$. We call $\mathcal{T}$ the *terminal set* of G, the vertices in $\mathcal{T}$ the *terminals*, and the other vertices *free vertices*. The path cover problem is a special case of the terminal path cover problem with $\mathcal{T} = \emptyset$. The terminal path cover problem can be applied to the applications of the path cover problem when some vertices corresponding to the applications are restricted to be terminals.

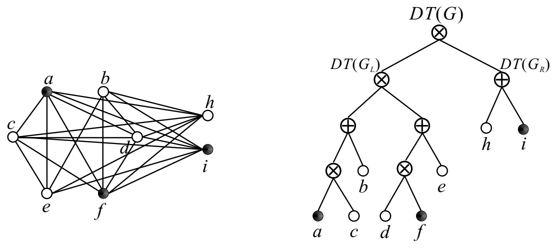
Cographs (also called complement-reducible graphs) are defined as the class of graphs formed from a single vertex under the closure of the operations of *union* and *complement*. Cographs were introduced by Lerchs [22], who studied their structural and algorithmic properties and enumerated the class. Names synonymous with cographs include D^*-graphs, P_4 restricted graphs, and Hereditary Dacey graphs. Several characterizations of cographs are known. For example, it is shown that G is a cograph if and only if G contains no P_4 (a path consisting of four vertices) as an induced subgraph [10]. Cographs have arisen in many disparate areas of mathematics and have been independently rediscovered by various researchers. Cographs can be recognized in linear time [11]. The class of cographs forms a subclass of distance-hereditary graphs [10, 11] and permutation graphs, and is a superclass of threshold graphs and complete-bipartite graphs. In addition, further properties and optimization problems in these graphs have been studied [1, 4, 5, 13, 14, 19, 21, 27, 30, 32, 33, 35, 37, 38].

Jung and Nicolai proposed linear time algorithms to solve the special cases of the terminal path cover problem with $|\mathcal{T}| = 1$ and $|\mathcal{T}| = 2$, respectively, on cographs [18, 29]. In this paper, we show that the terminal path cover problem whose $|\mathcal{T}|$ is not restricted to 2 on cographs can be solved in linear time.

2 Preliminaries

In this section, we define some notations and operations.

Definition 1. [10, 11] The class of cographs can be defined by the following recursive definition:
(1) A graph consisting of a single vertex and no edges is a cograph.
(2) If $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ are cographs, then the *union* G of G_L and G_R, denoted by $G = G_L \oplus G_R = (V_L \cup V_R, E_L \cup E_R)$, is a cograph. In this case, we say that G is formed from G_L and G_R by a *union operation*.
(3) If $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ are cographs, then the *joint* G of G_L and G_R, denoted by $G = G_L \otimes G_R = (V_L \cup V_R, E_L \cup E_R \cup \widehat{E})$, is a cograph, where $\widehat{E} = \{(u, v) | \forall u \in V_L \text{ and } \forall v \in V_R\}$. In this case, we say that G is formed from G_L and G_R by a *joint operation*.



(a) (b)

Figure 1: (a) A cograph G and a terminal set $\mathcal{T} = \{a, f, i\}$ of G, and (b) a decomposition tree $DT(G)$ of G, where full vertices indicate the terminals in G.

A cograph G can be represented by a rooted binary tree $DT(G)$, called a *decomposition tree* [7, 10]. The leaf nodes of $DT(G)$ represent the vertices of G. Each internal node of $DT(G)$ is labeled by either '$\oplus$' or '$\otimes$'. The cograph corresponding to a $\oplus$-labeled (resp. $\otimes$-labeled) node v in $DT(G)$ is obtained from the cographs corresponding to the children of v in $DT(G)$ by means of a *union* (resp. *joint*) operation. A decomposition tree of a cograph can be constructed as follows:

Definition 2. [7] The decomposition tree $DT(G)$ of a cograph G consisting of a single vertex v is a tree of one node labeled by v. If G is formed from G_L and G_R by a union (resp. joint) operation, then the root of the decomposition $DT(G)$ is a node labeled by $\oplus$ (resp. $\otimes$) with the roots of $DT(G_L)$ and $DT(G_R)$ being the children of the root of $DT(G)$, respectively.

Notice that the decomposition tree $DT(G)$ of a cograph G is a rooted and unordered binary tree. Exchanging the left and right children of an internal node in $DT(G)$ will be also a decomposition tree of G. For instance, given a cograph G shown in Figure 1(a), the decomposition tree $DT(G)$ of G is shown in Figure 1(b).

Theorem 2.1. [7, 10] *A decomposition tree $DT(G)$ of a cograph $G = (V, E)$ can be constructed in* $O(|V| + |E|)$ *time.*

A cograph is not connected if the root of its corresponding decomposition tree is a $\oplus$-labeled node. Hence, we assume that the root of the corresponding decomposition tree is a $\otimes$-labeled node.

In the rest of the paper, assume that $G = (V, E)$ is a cograph and is formed from G_L and G_R by either a union operation or a joint operation. We use V_L and V_R to denote the vertex sets of G_L and

G_R, respectively. In other words, $V = V_L \cup V_R$ and $V_L \cap V_R = \emptyset$. Notice that every vertex in V_L is adjacent to all vertices in V_R if G is formed from G_L and G_R by a joint operation.

Definition 3. A path of a cograph G is called *terminal path* if it visits at least two vertices and both of its end vertices are terminals. A path of G is called *semi-terminal path* if exactly one end vertex of it is a terminal. A path of G is called *non-terminal path* if neither of its end vertices is a terminal.

By the above definition, a semi-terminal path may consist of only one vertex which is a terminal, a non-terminal path may visit only one free vertex, and a terminal path must contain at least two vertices which are distinct terminals.

Definition 4. A terminal path cover $\mathcal{PC}$ of a cograph G with respect to $\mathcal{T}$ is called a (n, s, t)-*constrained path cover* if (1) $|\mathcal{PC}| = n + s + t$; (2) there are exactly n non-terminal paths in $\mathcal{PC}$; (3) there are exactly s semi-terminal paths in $\mathcal{PC}$, and (4) all other paths in $\mathcal{PC}$ are terminal paths.

A path in a terminal path cover of a cograph G is clearly either a non-terminal path, a semi-terminal path or a terminal path of G. Hence, the following proposition immediately holds:

Proposition 2.2. *Let G be a cograph and let $\mathcal{T}$ be a terminal set of G. Then, the following statements hold:*
(1)*$\mathcal{PC}$ is a terminal path cover of G with respect to $\mathcal{T}$ if and only if it is a constrained path cover of G.*
(2)*If G has a (n, s, t)-constrained path cover, then $|\mathcal{T}| = s + 2t$.*

Definition 5. Let $\mathcal{PC}$ be a (n, s, t)-constrained path cover of G. We say that $\mathcal{PC}$ is a *minimum constrained path cover* of G if $\pi(G, \mathcal{T}) = n + s + t$. Denote by $N_G(\mathcal{PC})$, $S_G(\mathcal{PC})$, and $T_G(\mathcal{PC})$, the subsets of $\mathcal{PC}$ containing all non-terminal paths, all semi-terminal paths, and all terminal paths in $\mathcal{PC}$, respectively. Define $f_G(\mathcal{PC}) = 2n + s$.

Definition 6. Let $\mathcal{P}$ be a set of vertex-disjoint paths in G. Denote by $F(\mathcal{P})$ the set of free vertices in $\mathcal{P}$. Note that $\mathcal{P}$ might not be a terminal path cover of G.

Let $\mathcal{P}$ be a set of vertex-disjoint paths with n non-terminal paths and s semi-terminal paths. By definition, $|F(\mathcal{P})| \geqslant n$. Given $\mathcal{P}$ and a number $\widehat{n}$, where $n \leqslant \widehat{n} \leqslant |F(\mathcal{P})|$, we can easily obtain a set $\widehat{\mathcal{P}}$ of vertex-disjoint paths with $\widehat{n}$ non-terminal paths and s semi-terminal paths by splitting paths in $\mathcal{P}$. Hence, the following proposition is clearly true:

Proposition 2.3. *For a set $\mathcal{P}$ of vertex-disjoint paths with n non-terminal paths and s semi-terminal paths, there exists a set $\widehat{\mathcal{P}}$ of vertex-disjoint paths with $\widehat{n}$ non-terminal paths and s semi-terminal paths such that $n \leqslant \widehat{n} \leqslant |F(\mathcal{P})|$.*

Definition 7. Let $\mathcal{PC}$ be a constrained path cover of a cograph G, P be a path in $\mathcal{PC}$, and let U be either V_L or V_R. A subpath P' of path P is called a U-subpath of P if P' visits vertices in U only. A U-subpath of P is U-*maximal* if it is not a proper subpath of any U-subpath of P. Denote by $U(\mathcal{PC})$, the set of all U-maximal subpaths of all paths in $\mathcal{PC}$.

Let G be a cograph formed from $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ by either a union or a joint operation and let $\mathcal{PC}$ be a constrained path cover of G. By definition, $V_L(\mathcal{PC})$ and $V_R(\mathcal{PC})$ are constrained path covers of G_L and G_R, respectively. It is easy to see that $V_L(\mathcal{PC}) \cup V_R(\mathcal{PC})$ is a constrained path cover of G and $\mathcal{PC} = V_L(\mathcal{PC}) \cup V_R(\mathcal{PC})$ if $G = G_L \oplus G_R$. Assume that $G = G_L \otimes G_R$ below. Consider the process to generate $\mathcal{PC}$ by adding edges between $V_L(\mathcal{PC})$ and $V_R(\mathcal{PC})$, one edge at a time, to the current constrained path cover starting from $V_L(\mathcal{PC}) \cup V_R(\mathcal{PC})$. The process is repeated until $\mathcal{PC}$ is obtained. Observe that, after adding one edge, the number of paths in the resulting constrained path cover is decreased by one. The number of edges adding to $V_L(\mathcal{PC}) \cup V_R(\mathcal{PC})$ is clearly at most $\min\{f_{G_L}(V_L(\mathcal{PC})), f_{G_R}(V_R(\mathcal{PC}))\}$. Therefore, we have the following proposition:

Proposition 2.4. *Assume that G is a cograph formed from $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ by either a union or a joint operation and that $\mathcal{PC}$ is a constrained path cover of G. Then, $|V_L(\mathcal{PC})| + |V_R(\mathcal{PC})| \geqslant |\mathcal{PC}| \geqslant |V_L(\mathcal{PC})| + |V_R(\mathcal{PC})| - \min\{f_{G_L}(V_L(\mathcal{PC})), f_{G_R}(V_R(\mathcal{PC}))\}$.*

For instance, let $G = G_L \otimes G_R$ be a cograph shown in Figure 1, $\mathcal{T} = \{a, f, i\}$ be a terminal set of G, and let $\mathcal{PC} = \{P_1 = a \to h \to c \to e \to b \to d \to f, P_2 = i\}$ be a constrained path cover of G, where P_1 and P_2 are terminal path and semi-terminal path of G, respectively. Then, $\mathcal{PC}$ is a $(0, 1, 1)$-constrained path cover of G, $V_L(\mathcal{PC}) = \{a, c \to e \to b \to d \to f\}$ is a $(0, 2, 0)$-constrained path cover of G_L, and $V_R(\mathcal{PC}) = \{h, i\}$ is a $(1, 1, 0)$-constrained path cover of G_R. By definition, $f_{G_L}(V_L(\mathcal{PC})) = 2$ and $f_{G_R}(V_R(\mathcal{PC})) =$

3. Hence, $\min\{f_{G_L}(V_L(\mathcal{PC})), f_{G_R}(V_R(\mathcal{PC}))\} = 2$, $|V_L(\mathcal{PC})| + |V_R(\mathcal{PC})| = 4$, and $|V_L(\mathcal{PC})| + |V_R(\mathcal{PC})| - \min\{f_{G_L}(V_L(\mathcal{PC})), f_{G_R}(V_R(\mathcal{PC}))\} = 2$. By Proposition 2.4, $4 \geqslant |\mathcal{PC}| = 2 \geqslant 2$.

Let P_v^u denote a path of a graph with $start(P_v^u) = u$ and $end(P_v^u) = v$. We allow that $u = v$ only in the case that P_v^u contains exactly one vertex. For two vertex-disjoint paths P_v^u and P_y^x such that v is adjacent to x, let $P_v^u \circ P_y^x$ denote the *concatenation* of P_v^u with P_y^x; that is, $P_y^u = P_v^u \circ P_y^x$ and P_y^u visits every vertex of P_v^u and P_y^x exactly once.

We then define some operations on two distinct sets of vertex-disjoint paths in the following.

Definition 8. ($\rhd$ operation) Let $\mathcal{P} = \{P_{v_1}^{u_1}, P_{v_2}^{u_2}, \cdots, P_{v_p}^{u_p}\}$ and $\mathcal{Q}=\{P_{y_1}^{x_1}, P_{y_2}^{x_2}, \cdots, P_{y_q}^{x_q}\}$ be two sets of vertex-disjoint paths such that $p \geqslant q$, v_i is adjacent to x_i, and both v_i and x_i are not terminals for $i \leqslant q$. For an integer $k \leqslant q$, define $\rhd_k$ to be the operation on $\mathcal{P}$ and $\mathcal{Q}$, denoted by $\mathcal{P} \rhd_k \mathcal{Q}$, that constructs a set of paths $\{P_{v_1}^{u_1} \circ P_{y_1}^{x_1}, P_{v_2}^{u_2} \circ P_{y_2}^{x_2}, \cdots, P_{v_k}^{u_k} \circ P_{y_k}^{x_k}, P_{v_{k+1}}^{u_{k+1}}, \cdots, P_{v_p}^{u_p}, P_{y_{k+1}}^{x_{k+1}}, \cdots, P_{y_q}^{x_q}\}$. Figure 2(a) depicts the operation.

Definition 9. ($\bowtie$ operation) Let $\mathcal{P} =\{P_{v_1}^{u_1}, P_{v_2}^{u_2}, \cdots, P_{v_p}^{u_p}\}$ and $\mathcal{Q} =\{P_{y_1}^{x_1}, P_{y_2}^{x_2}, \cdots, P_{y_q}^{x_q}\}$ be two sets of vertex-disjoint paths such that $p > q$, both end vertices of each path in $\mathcal{P}$ are adjacent to all end vertices of paths in $\mathcal{Q}$, and all end vertices of paths in $\mathcal{P}$ and $\mathcal{Q}$ are not terminals. For an integer $k \leqslant q$, define $\bowtie_k$ to be the operation on $\mathcal{P}$ and $\mathcal{Q}$, denoted by $\mathcal{P} \bowtie_k \mathcal{Q}$, that generates a set of paths $\{P_{v_1}^{u_1} \circ P_{y_1}^{x_1} \circ P_{v_2}^{u_2} \circ P_{y_2}^{x_2} \circ \cdots \circ P_{v_k}^{u_k} \circ P_{y_k}^{x_k} \circ P_{v_{k+1}}^{u_{k+1}}, P_{v_{k+2}}^{u_{k+2}}, \cdots, P_{v_p}^{u_p}, P_{y_{k+1}}^{x_{k+1}}, \cdots, P_{y_q}^{x_q}\}$. The operation is revealed in Figure 2(b).

Let $\mathcal{P}$ and $\mathcal{Q}$ be defined as the above definition except that $p \geqslant q \geqslant k+1$. For simplicity, we will use $\mathcal{P}(\bowtie_k, \rhd_1)\mathcal{Q}$ to denote the set of paths $\{P_{v_1}^{u_1} \circ P_{y_1}^{x_1} \circ P_{v_2}^{u_2} \circ P_{y_2}^{x_2} \circ \cdots \circ P_{v_k}^{u_k} \circ P_{y_k}^{x_k} \circ P_{v_{k+1}}^{u_{k+1}} \circ P_{y_{k+1}}^{x_{k+1}}, P_{v_{k+2}}^{u_{k+2}}, \cdots, P_{v_p}^{u_p}, P_{y_{k+2}}^{x_{k+2}}, \cdots, P_{y_q}^{x_q}\}$

Definition 10. ($\ltimes$ operation) Let $\mathcal{P} =\{P_{v_1}^{u_1}, P_{v_2}^{u_2}, \cdots, P_{v_p}^{u_p}\}$ and $\mathcal{Q} =\{P_{y_1}^{x_1}, P_{y_2}^{x_2}, \cdots, P_{y_q}^{x_q}\}$ be two sets of vertex-disjoint paths such that both end vertices of each path in $\mathcal{P}$ are adjacent to all end vertices of paths in $\mathcal{Q}$, all end vertices of paths in $\mathcal{P}$ are not terminals, and x_i is a terminal and y_i is a free vertex for $1 \leqslant i \leqslant q$. For an integer $k \leqslant \min\{p, \lfloor \frac{q}{2} \rfloor\}$, define $\ltimes_k$ to be the operation on $\mathcal{P}$ and $\mathcal{Q}$, denoted by $\mathcal{P} \ltimes_k \mathcal{Q}$, that generates a set of paths $\{P_{y_1}^{x_1} \circ P_{v_1}^{u_1} \circ P_{x_2}^{y_2}, P_{y_3}^{x_3} \circ P_{v_2}^{u_2} \circ P_{x_4}^{y_4}, \cdots, P_{y_{2k-1}}^{x_{2k-1}} \circ P_{v_k}^{u_k} \circ P_{x_{2k}}^{y_{2k}}, P_{v_{k+1}}^{u_{k+1}}, P_{v_{k+2}}^{u_{k+2}}, \cdots, P_{v_p}^{u_p}, P_{y_{2k+1}}^{x_{2k+1}}, \cdots, P_{y_q}^{x_q}\}$. Figure 2(c) shows the operation.

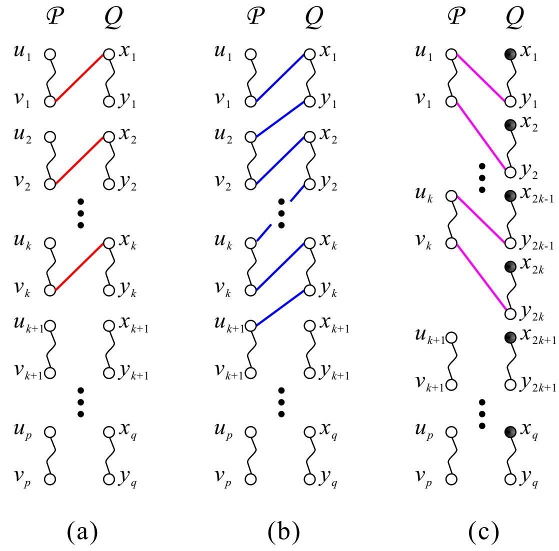


Figure 2: Operations defined on two sets $\mathcal{P}$ and $\mathcal{Q}$ of vertex-disjoint paths, where (a) $\mathcal{P} \rhd_k \mathcal{Q}$, (b) $\mathcal{P} \bowtie_k \mathcal{Q}$, and (c) $\mathcal{P} \ltimes_k \mathcal{Q}$, where full vertices indicate terminals.

3 The Terminal Path Cover Problem in Cographs

In [9, 10, 18, 23], they showed that the path cover problem on cographs can be solved in linear time. Note that the path cover problem is a special case of the terminal path cover problem where the terminal set of the input graph is empty.

Lemma 3.1. [9, 10, 18, 23] *Let $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ be two cographs. Then, the following two statements hold:*
(1) If $G = G_L \oplus G_R$, then $\pi(G) = \pi(G_L) + \pi(G_R)$.
(2) If $G = G_L \otimes G_R$, then $\pi(G) = \max\{1, \pi(G_L) - |V_R|, \pi(G_R) - |V_L|\}$.

In addition, we can easily see that the following lemma holds:

Lemma 3.2. *Let G_L and G_R be two cographs with terminal sets $\mathcal{T}_L$ and $\mathcal{T}_R$, respectively. If $G = G_L \oplus G_R$, then $\pi(G, \mathcal{T}_L \cup \mathcal{T}_R) = \pi(G_L, \mathcal{T}_L) + \pi(G_R, \mathcal{T}_R)$.*

By the above two lemmas, we can easily compute $\pi(G, \mathcal{T})$ if $G = G_L \oplus G_R$ or $\mathcal{T} = \mathcal{T}_L \cup \mathcal{T}_R = \emptyset$. In the rest of the paper, we focus on computing $\pi(G, \mathcal{T})$ under that $G = G_L \otimes G_R$ and $\mathcal{T} \neq \emptyset$.

We first find the relation between $f_G(\mathcal{PC})$ and $f_G(\widehat{\mathcal{PC}})$, where $\mathcal{PC}$ and $\widehat{\mathcal{PC}}$ are constrained path covers of a cograph G, $\mathcal{PC}$ is a minimum constrained path cover of G but $\widehat{\mathcal{PC}}$ is not minimum. Recall that $f_G(\mathcal{PC}) = 2n + s$ for a (n, s, t)-constrained path cover $\mathcal{PC}$ of G.

Lemma 3.3. *Let G be a cograph with terminal set $\mathcal{T}$. Suppose that $\mathcal{PC}_1$ and $\mathcal{PC}_2$ are minimum constrained path covers of G and that $\widehat{\mathcal{PC}}$ is a constrained path cover of G which is not minimum. Then, $f_G(\mathcal{PC}_1) = f_G(\mathcal{PC}_2)$ and $f_G(\mathcal{PC}_1) < f_G(\widehat{\mathcal{PC}})$.*

Proof. Assume that $\mathcal{PC}_1$, $\mathcal{PC}_2$, and $\widehat{\mathcal{PC}}$ are (n_1, s_1, t_1)-constrained, (n_2, s_2, t_2)-constrained, and $(\widehat{n}, \widehat{s}, \widehat{t})$-constrained path covers of G, respectively. By definition, $f_G(\mathcal{PC}_1) = 2n_1 + s_1$, $f_G(\mathcal{PC}_2) = 2n_2 + s_2$, and $f_G(\widehat{\mathcal{PC}}) = 2\widehat{n} + \widehat{s}$. We will prove the lemma by showing that $2n_1 + s_1 = 2n_2 + s_2$ and $2n_1 + s_1 < 2\widehat{n} + \widehat{s}$. By Proposition 2.2, we obtain that

$$|\mathcal{T}| = s_1 + 2t_1 = s_2 + 2t_2 = \widehat{s} + 2\widehat{t}. \tag{1}$$

Since both $\mathcal{PC}_1$ and $\mathcal{PC}_2$ are minimum constrained path covers of G and $\widehat{\mathcal{PC}}$ is not a minimum constrained path cover of G, we get that

$$n_1 + s_1 + t_1 = n_2 + s_2 + t_2, \tag{2}$$

$$n_1 + s_1 + t_1 < \widehat{n} + \widehat{s} + \widehat{t}. \tag{3}$$

By Eq. (2), $n_1 = n_2 + s_2 + t_2 - s_1 - t_1$. Hence, $2n_1 + s_1 = 2(n_2 + s_2 + t_2 - s_1 - t_1) + s_1 = 2n_2 + s_2 + (s_2 + 2t_2 - s_1 - 2t_1)$. By Eq. (1), $s_2 + 2t_2 - s_1 - 2t_1 = 0$. Hence, $2n_1 + s_1 = 2n_2 + s_2$. By Eq. (3), $n_1 + s_1 < \widehat{n} + \widehat{s} + \widehat{t} - t_1$. Hence, $2n_1 + s_1 < 2(\widehat{n} + \widehat{s} + \widehat{t} - t_1) - s_1 = 2\widehat{n} + \widehat{s} + (\widehat{s} + 2\widehat{t} - 2t_1 - s_1)$. By Eq. (1), we have that $\widehat{s} + 2\widehat{t} - 2t_1 - s_1 = 0$. Hence, $2n_1 + s_1 < 2\widehat{n} + \widehat{s}$. □

Corollary 3.4. *Let $G = (V, E)$ be a cograph with terminal set $\mathcal{T}$. Suppose that $\mathcal{PC}$ is a minimum constrained path cover of G and that $\widehat{\mathcal{PC}}$ is a constrained path cover of G. Then, $f_G(\mathcal{PC}) \leqslant f_G(\widehat{\mathcal{PC}}) \leqslant 2|V \setminus \mathcal{T}| + |\mathcal{T}|$.*

Proof. Clearly, $f_G(\widehat{\mathcal{PC}}) \leqslant 2|V \setminus \mathcal{T}| + |\mathcal{T}|$. By Lemma 3.3, $f_G(\mathcal{PC}) \leqslant f_G(\widehat{\mathcal{PC}})$. Hence the corollary holds. □

Without of loss generality, we assume the following statements hold for all lemmas in the rest of the paper.

Assumption 1. (1) $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ are two cographs with terminal sets $\mathcal{T}_L$ and $\mathcal{T}_R$, respectively; (2) $G = G_L \otimes G_R$ is a cograph with terminal set $\mathcal{T} = \mathcal{T}_L \cup \mathcal{T}_R$; (3) $|\mathcal{T}| \geqslant 1$ and $|\mathcal{T}_L| \geqslant |\mathcal{T}_R|$, and (4) $\mathcal{PC}_L$ (resp. $\mathcal{PC}_R$) is a minimum and (n_L, s_L, t_L)-constrained (resp. (n_R, s_R, t_R)-constrained) path cover of G_L (resp. G_R) satisfying that there exists no minimum constrained path cover of G_L (resp. G_R) with semi-terminal path if $s_L = 0$ (resp. $s_R = 0$).

In the following, we will show that given $\pi(G_L, \mathcal{T}_L)$ and $\pi(G_R, \mathcal{T}_R)$, $\pi(G, \mathcal{T})$ can be computed in constant time. We first obtain the lower bound of $\pi(G, \mathcal{T})$ in the following lemma:

Lemma 3.5. $\pi(G, \mathcal{T}) \geqslant \max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\}$.

Proof. Clearly, $\pi(G, \mathcal{T}) \geqslant \lceil \frac{|\mathcal{T}|}{2} \rceil$. Now, we will prove that $\pi(G, \mathcal{T}) \geqslant \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. Assume by contradiction that $\pi(G, \mathcal{T}) < \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. Then, G has a minimum constrained path cover $\mathcal{PC}$ of size p with $p < \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. Let $G^* = G \setminus (V_R \setminus \mathcal{T}_R)$. Clearly, $\pi(G^*, \mathcal{T}) \geqslant \pi(G_L, \mathcal{T}_L)$. Consider removing from $\mathcal{PC}$ all vertices in $V_R \setminus \mathcal{T}_R$. What results is a set $\mathcal{PC}^*$ of vertex-disjoint paths which is clearly a constrained path cover of G^* with respect to $\mathcal{T}$. Since the removal of a vertex in $V_R \setminus \mathcal{T}_R$ will increase the number of paths by at most one, we can obtain a constrained path cover of G^* of size at most $p + |V_R \setminus \mathcal{T}_R|$ and hence, $|\mathcal{PC}^*| \leqslant p + |V_R \setminus \mathcal{T}_R|$. Therefore, $p + |V_R \setminus \mathcal{T}_R| \geqslant |\mathcal{PC}^*| \geqslant \pi(G^*, \mathcal{T})$. Since $p + |V_R \setminus \mathcal{T}_R| \geqslant \pi(G^*, \mathcal{T})$ and $\pi(G^*, \mathcal{T}) \geqslant \pi(G_L, \mathcal{T}_L)$, we get that $p \geqslant \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. It contradicts that $p < \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. Thus, $\pi(G, \mathcal{T}) \geqslant \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$.

Using similar arguments as those for proving that $\pi(G, \mathcal{T}) \geqslant \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$, we can prove that $\pi(G, \mathcal{T}) \geqslant \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|$. Hence, $\pi(G, \mathcal{T}) \geqslant \max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\}$. □

Next, we would like to find the upper bound of $\pi(G, \mathcal{T})$. In other words, we show that $\pi(G, \mathcal{T}) \leqslant \max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\}$ except the special case of $s_L = |\mathcal{T}_R| = 0$ and $n_L = |V_R|$. The following lemma shows that $\pi(G, \mathcal{T}) = \frac{|\mathcal{T}|}{2} + 1$ if $s_L = |\mathcal{T}_R| = 0$ and $n_L = |V_R|$:

Lemma 3.6. *Assume that $s_L = |\mathcal{T}_R| = 0$ and $n_L = |V_R|$. Then, $\pi(G, \mathcal{T}) = \frac{|\mathcal{T}|}{2} + 1$.*

Proof. By Assumption 1, $|\mathcal{T}| = |\mathcal{T}_L| \geqslant 1$. By Proposition 2.2, $|\mathcal{T}_L| = s_L + 2t_L$. Since $s_L = 0$, we have that $t_L = \frac{|\mathcal{T}|}{2} > 0$.

We first prove that $\pi(G, \mathcal{T}) \geqslant \frac{|\mathcal{T}|}{2} + 1$. Assume by contradiction that $\pi(G, \mathcal{T}) = \frac{|\mathcal{T}|}{2}$. Suppose that $\widehat{\mathcal{PC}}$ is a minimum constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$. Then, $V_L(\widehat{\mathcal{PC}})$ and $V_R(\widehat{\mathcal{PC}})$ are $(\widehat{n}_L, \widehat{s}_L, \widehat{t}_L)$-constrained and $(\widehat{n}_R, \widehat{s}_R, \widehat{t}_R)$-constrained path covers of G_L and G_R, respectively. Since $|\mathcal{T}_R| = 0$, $\widehat{s}_R = \widehat{t}_R = 0$. By definition, we have that

$$n_R \leqslant \widehat{n}_R \leqslant |V_R| = n_L. \tag{4}$$

Since $|\widehat{\mathcal{PC}}| = \frac{|\mathcal{T}|}{2}$, we have that $\widehat{\mathcal{PC}} = T_G(\widehat{\mathcal{PC}})$. There are two types of paths in $\widehat{\mathcal{PC}}$: type-1 paths are those paths visiting vertices in V_L only; and type-2 paths are those paths visiting vertices in both V_L and V_R with both end vertices in $\mathcal{T}_L$. The number of paths of type-1 is equal to $\widehat{t}_L$. Suppose that $f_{G_R}(V_R(\widehat{\mathcal{PC}})) < f_{G_L}(V_L(\widehat{\mathcal{PC}}))$. Then, $2\widehat{n}_R < 2\widehat{n}_L + \widehat{s}_L$. By Proposition 2.4, $|\widehat{\mathcal{PC}}| = \frac{|\mathcal{T}_L|}{2} = \frac{\widehat{s}_L+2\widehat{t}_L}{2} = \frac{\widehat{s}_L}{2} + \widehat{t}_L \geqslant |V_L(\widehat{\mathcal{PC}})| + |V_R(\widehat{\mathcal{PC}})| - f_{G_R}(V_R(\widehat{\mathcal{PC}})) = \widehat{n}_L + \widehat{s}_L + \widehat{t}_L - \widehat{n}_R - 2\widehat{n}_R$. Hence, $2\widehat{n}_R \geqslant 2\widehat{n}_L + \widehat{s}_L$ and a contradiction occurs. On the other hand, suppose that $f_{G_L}(V_L(\widehat{\mathcal{PC}})) < f_{G_R}(V_R(\widehat{\mathcal{PC}}))$. Then, $2\widehat{n}_L + \widehat{s}_L < 2\widehat{n}_R$. By Proposition 2.4, $|\widehat{\mathcal{PC}}| = \frac{\widehat{s}_L}{2} + \widehat{t}_L \geqslant |V_L(\widehat{\mathcal{PC}})| + |V_R(\widehat{\mathcal{PC}})| - f_{G_L}(V_L(\widehat{\mathcal{PC}})) = \widehat{n}_L + \widehat{s}_L + \widehat{t}_L + \widehat{n}_R - 2\widehat{n}_L - \widehat{s}_L$. Hence, $2\widehat{n}_L + \widehat{s}_L \geqslant 2\widehat{n}_R$ and a contradiction occurs too. Therefore, $f_{G_L}(V_L(\widehat{\mathcal{PC}})) = f_{G_R}(V_R(\widehat{\mathcal{PC}}))$ and we get that

$$2\widehat{n}_L + \widehat{s}_L = 2\widehat{n}_R. \tag{5}$$

A type-2 path P is clearly a terminal path of G with both end vertices in $\mathcal{T}_L$ since $|\widehat{\mathcal{PC}}| = \frac{|\mathcal{T}_L|}{2}$. Clearly, P contains exactly two semi-terminal paths of G_L. Since $V_R \neq \emptyset$, there exists at least one type-2 path in $\widehat{\mathcal{PC}}$. Thus, $\widehat{s}_L \neq 0$. By Assumption 1, $V_L(\widehat{\mathcal{PC}})$ is not a minimum constrained path cover of G_L. By Lemma 3.3, we obtain that

$$2n_L < 2\widehat{n}_L + \widehat{s}_L. \tag{6}$$

Combining Eqs. (4)–(6), we get that $2n_L < 2\widehat{n}_L + \widehat{s}_L = 2\widehat{n}_R \leqslant 2|V_R| = 2n_L$ and hence, a contradiction occurs. Hence, $\pi(G, \mathcal{T}) \neq \frac{|\mathcal{T}|}{2}$, i.e., $\pi(G, \mathcal{T}) \geqslant \frac{|\mathcal{T}|}{2} + 1$.

Next we construct a constrained path cover of G of size $\frac{|\mathcal{T}|}{2} + 1$ from $\mathcal{PC}_L$ and V_R. Let $\mathcal{P} = N_{G_L}(\mathcal{PC}_L)(\bowtie_{|V_R|-1}, \rhd_1)V_R$. Then, $\mathcal{P}$ contains only a non-terminal path P_n of G with one end vertex in V_L and the other in V_R. Let $P_t = v_t \circ P_s$ be a path in $T_{G_L}(\mathcal{PC}_L)$, where $v_t \in \mathcal{T}_L$. Let $\mathcal{PC}_t = T_{G_L}(\mathcal{PC}_L) \setminus \{P_t\}$, $\mathcal{PC}_s = \{P_s, v_t \circ P_n\}$, and let $\mathcal{PC} = \mathcal{PC}_s \cup \mathcal{PC}_t$. Then, $\mathcal{PC}$ is a $(0, 2, \frac{|\mathcal{T}|}{2} - 1)$-constrained path cover of G. Thus, $\pi(G, \mathcal{T}) \leqslant \frac{|\mathcal{T}|}{2} + 1$.

It follows immediately from the above arguments that $\pi(G, \mathcal{T}) = \frac{|\mathcal{T}|}{2} + 1$. □

For the above case of $s_L = |\mathcal{T}_R| = 0$ and $n_L = |V_R|$, we construct a minimum constrained path cover of G with two semi-terminal paths. We then make the following assumption:

Assumption 2. The equations of $s_L = |\mathcal{T}_R| = 0$ and $n_L = |V_R|$ do not hold at the same time.

The above assumption implies that $n_L \neq |V_R|$ if $s_L = |\mathcal{T}_R| = 0$.

Let $\widehat{\mathcal{PC}}_L$ and $\widehat{\mathcal{PC}}_R$ be constrained path covers of G_L and G_R, respectively. By Corollary 3.4, $2n_L + s_L = f_{G_L}(\mathcal{PC}_L) \leqslant f_{G_L}(\widehat{\mathcal{PC}}_L) \leqslant 2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L|$ and $2n_R + s_R = f_{G_R}(\mathcal{PC}_R) \leqslant f_{G_R}(\widehat{\mathcal{PC}}_R) \leqslant 2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R|$. Considering the relation between $f_{G_L}(\widehat{\mathcal{PC}}_L)$ and $f_{G_R}(\widehat{\mathcal{PC}}_R)$, we have the following four cases:

Case I: $2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L| < 2n_R + s_R$;
Case II: $2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R| < 2n_L + s_L$;
Case III: $2n_L + s_L \leqslant 2n_R + s_R \leqslant 2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L|$;
Case IV: $2n_R + s_R \leqslant 2n_L + s_L \leqslant 2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R|$.

For Cases I and II, we will construct constrained path covers of G of sizes $\pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|$ and $\pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$, respectively. For the other cases, we will construct a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$. Note that the conditions appeared in Lemma 3.6 hold only for Case III or Case IV. Following Lemma 3.5 and Lemma 3.7, we obtain that $\pi(G, \mathcal{T}) = \max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\}$ if Assumption 2 holds.

Lemma 3.7. *The following statements hold:*
(1)*If* $2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L| < 2n_R + s_R$, *then* $\max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\} = \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|$.
(2)*If* $2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R| < 2n_L + s_L$, *then* $\max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\} = \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$.

Proof. We prove Statement (1) by showing that $2(\pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|) > |\mathcal{T}|$ and $\pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L| > \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$.

We first prove that $2(\pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|) > |\mathcal{T}|$. By definition, we have that

$$\pi(G_R, \mathcal{T}_R) = n_R + s_R + t_R. \tag{7}$$

By assumption of the lemma, we obtain that

$$2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L| < 2n_R + s_R. \tag{8}$$

By Proposition 2.2, $|\mathcal{T}_L| = s_L + 2t_L$ and $|\mathcal{T}_R| = s_R + 2t_R$. Combining Eqs. (7) and (8), we get that

$2(\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|)=2(n_R+s_R+t_R-|V_L\setminus\mathcal{T}_L|)=2n_R+s_R+|\mathcal{T}_R|-2|V_L\setminus\mathcal{T}_L|>(2|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|)+|\mathcal{T}_R|-2|V_L\setminus\mathcal{T}_L|=|\mathcal{T}_L|+|\mathcal{T}_R|=|\mathcal{T}|$.

Next, we prove that $\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|>\pi(G_L,\mathcal{T}_L)-|V_R\setminus\mathcal{T}_R|$. By definition, we have that

$$\pi(G_L,\mathcal{T}_L)=n_L+s_L+t_L\leqslant|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|. \quad (9)$$

Combining Eqs. (8) and (9), we obtain that

$$\pi(G_L,\mathcal{T}_L)+|V_L\setminus\mathcal{T}_L|<2n_R+s_R. \quad (10)$$

By Corollary 3.4, $2n_R+s_R\leqslant 2|V_R\setminus\mathcal{T}_R|+|\mathcal{T}_R|$. By Eq. (8), we obtain that

$$2|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|<2|V_R\setminus\mathcal{T}_R|+|\mathcal{T}_R|. \quad (11)$$

Combining Eqs. (10) and (11), we get that $2(\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|-\pi(G_L,\mathcal{T}_L)+|V_R\setminus\mathcal{T}_R|)=2(n_R+s_R+t_R-|V_L\setminus\mathcal{T}_L|-\pi(G_L,\mathcal{T}_L)+|V_R\setminus\mathcal{T}_R|)=2n_R+s_R+|\mathcal{T}_R|-2|V_L\setminus\mathcal{T}_L|-2\pi(G_L,\mathcal{T}_L)+2|V_R\setminus\mathcal{T}_R|>(\pi(G_L,\mathcal{T}_L)+|V_L\setminus\mathcal{T}_L|)+|\mathcal{T}_R|-2|V_L\setminus\mathcal{T}_L|-2\pi(G_L,\mathcal{T}_L)+2|V_R\setminus\mathcal{T}_R|=-\pi(G_L,\mathcal{T}_L)-|V_L\setminus\mathcal{T}_L|+2|V_R\setminus\mathcal{T}_R|+|\mathcal{T}_R|>-\pi(G_L,\mathcal{T}_L)-|V_L\setminus\mathcal{T}_L|+(2|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|)=|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|-\pi(G_L,\mathcal{T}_L)\geqslant 0$. Hence, $\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|>\pi(G_L,\mathcal{T}_L)-|V_R\setminus\mathcal{T}_R|$.

By symmetry, Statement (2) can be proved by using arguments similar to those used in proving Statement (1). □

Lemma 3.8. *If* $2|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|<2n_R+s_R$, *then* $\pi(G,\mathcal{T})=\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$.

Proof. We will prove this lemma by showing how to construct a constrained path cover of G of size $\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$. By Assumption 1, $|\mathcal{T}_L|\geqslant|\mathcal{T}_R|=s_R+2t_R$. By assumption of the lemma, $2n_R>2|V_L\setminus\mathcal{T}_L|+|\mathcal{T}_L|-s_R$ and hence, $n_R\geqslant|V_L\setminus\mathcal{T}_L|+\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor+1$. We then construct from V_L and $\mathcal{PC}_R$ a constrained path cover of G of size $\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$ through the following procedure:

1. partition $\mathcal{T}_L$ into two sets $\mathcal{T}_L^\alpha$ and $\mathcal{T}_L^\beta$ such that $|\mathcal{T}_L^\alpha|=s_R$ and $|\mathcal{T}_L^\beta|=|\mathcal{T}_L|-s_R$;
2. $\mathcal{PC}_{t_1}=\mathcal{T}_L^\alpha\rhd_{s_R}S_{G_R}(\mathcal{PC}_R)$;
3. $\mathcal{P}_N=N_{G_R}(\mathcal{PC}_R)\bowtie_{|V_L\setminus\mathcal{T}_L|}(V_L\setminus\mathcal{T}_L)$, where $|\mathcal{P}_N|=n_R-|V_L\setminus\mathcal{T}_L|\geqslant\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor+1$ and every path in $\mathcal{P}_N$ is a non-terminal path of G with both end vertices in V_R;
4. partition $\mathcal{P}_N$ into two sets $\mathcal{P}_N^\beta$ and $\mathcal{P}_N^\gamma$ such that $|\mathcal{P}_N^\beta|=\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor$ and $|\mathcal{P}_N^\gamma|=n_R-|V_L\setminus\mathcal{T}_L|-\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor$, where $|\mathcal{P}_N^\gamma|\geqslant 1$;
5. $\mathcal{PC}_{t_2}=\mathcal{P}_N^\beta\ltimes_{\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor}\mathcal{T}_L^\beta$, where there are exactly $\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor$ terminal paths in $\mathcal{PC}_{t_2}$;
6. **if** $|\mathcal{T}_L|-s_R$ is odd, i.e., there exists an isolated terminal v_t in $\mathcal{PC}_{t_2}$, **then**
7. let $\widehat{P}$ be a non-terminal path in $\mathcal{P}_N^\gamma$;
8. $\mathcal{PC}_{t_2}=\mathcal{PC}_{t_2}\setminus\{v_t\}$; $\mathcal{PC}_n=\mathcal{P}_N^\gamma\setminus\{\widehat{P}\}$; $\mathcal{PC}_s=\{v_t\circ\widehat{P}\}$;
9. $\mathcal{PC}_t=\mathcal{PC}_{t_1}\cup\mathcal{PC}_{t_2}\cup T_{G_R}(\mathcal{PC}_R)$;
10. **else** /* $|\mathcal{T}_L|-s_R$ is even*/
11. let P_t be a terminal path in $\mathcal{PC}_{t_1}\cup\mathcal{PC}_{t_2}$ such that $P_t=v_t\circ P_{s_1}$ and $v_t\in\mathcal{T}_L$;
12. let P_n be a non-terminal path in $\mathcal{P}_N^\gamma$;
13. $\mathcal{PC}_n=\mathcal{P}_N^\gamma\setminus\{P_n\}$; $\mathcal{PC}_s=\{P_{s_1},v_t\circ P_n\}$, and $\mathcal{PC}_t=\mathcal{PC}_{t_1}\cup\mathcal{PC}_{t_2}\cup T_{G_R}(\mathcal{PC}_R)\setminus\{P_t\}$;
14. $\mathcal{PC}=\mathcal{PC}_n\cup\mathcal{PC}_s\cup\mathcal{PC}_t$.

Following the above procedure, we construct a $(n_R-|V_L\setminus\mathcal{T}_L|-\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor-1,1,s_R+t_R+\lfloor\frac{|\mathcal{T}_L|-s_R}{2}\rfloor)$-constrained path cover $\mathcal{PC}$ of G if $|\mathcal{T}_L|-s_R$ is odd; and a $(n_R-|V_L\setminus\mathcal{T}_L|-\frac{|\mathcal{T}_L|-s_R}{2}-1,2,s_R+t_R+\frac{|\mathcal{T}_L|-s_R}{2}-1)$-constrained path cover of G otherwise. Hence, we can construct from V_L and $\mathcal{PC}_R$ a constrained path cover $\mathcal{PC}$ of G of size $\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$ with at least a semi-terminal path. That is, $\pi(G,\mathcal{T})\leqslant\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$. By Lemma 3.5 and Statement (1) of Lemma 3.7, we get that $\pi(G,\mathcal{T})\geqslant\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$. Hence, $\pi(G,\mathcal{T})=\pi(G_R,\mathcal{T}_R)-|V_L\setminus\mathcal{T}_L|$. □

Lemma 3.9. *If* $2|V_R\setminus\mathcal{T}_R|+|\mathcal{T}_R|<2n_L+s_L$, *then* $\pi(G,\mathcal{T})=\pi(G_L,\mathcal{T}_L)-|V_R\setminus\mathcal{T}_R|$.

Proof. By Assumption 1, $|\mathcal{T}_L|\geqslant|\mathcal{T}_R|$. By assumption of the lemma, $2n_L+s_L>2|V_R\setminus\mathcal{T}_R|+|\mathcal{T}_R|$ and hence, $n_L\geqslant|V_R\setminus\mathcal{T}_R|+\lfloor\frac{|\mathcal{T}_R|-s_L}{2}\rfloor+1$. Suppose that $|\mathcal{T}_R|>s_L$. By using arguments similar to those used in proving Lemma 3.8, we can construct from V_R and $\mathcal{PC}_L$ a constrained path cover $\mathcal{PC}$ of G of size $\pi(G_L,\mathcal{T}_L)-|V_R\setminus\mathcal{T}_R|$ such that $\mathcal{PC}$ contains at least one semi-terminal path. In the following, suppose that $|\mathcal{T}_R|\leqslant s_L$. Then, $n_L\geqslant|V_R\setminus\mathcal{T}_R|-\lfloor\frac{s_L-|\mathcal{T}_R|}{2}\rfloor+1$. For the case of $|V_R\setminus\mathcal{T}_R|<\lfloor\frac{s_L-|\mathcal{T}_R|}{2}\rfloor$, $n_L=0$ and we can easily obtain a minimum constrained path cover of G with semi-terminal paths as follows: (1) partition $S_{G_L}(\mathcal{PC}_L)$ into two sets S_L^α and S_L^β such that $|S_L^\alpha|=|\mathcal{T}_R|$ and $|S_L^\beta|=s_L-|\mathcal{T}_R|$; (2) $\mathcal{P}_T=S_L^\alpha\rhd_{|\mathcal{T}_R|}\mathcal{T}_R$; (3) $\mathcal{P}_S=(V_R\setminus\mathcal{T}_R)\ltimes_{|V_R\setminus\mathcal{T}_R|}S_L^\beta$, and (4) $\mathcal{PC}=\mathcal{P}_T\cup\mathcal{P}_S\cup N_{G_L}(\mathcal{PC}_L)$. Note that there exists at least one semi-terminal path in $\mathcal{P}_S$ since $|V_R\setminus\mathcal{T}_R|<\lfloor\frac{s_L-|\mathcal{T}_R|}{2}\rfloor$. In the following, we only consider the case that $|V_R\setminus\mathcal{T}_R|\geqslant\lfloor\frac{s_L-|\mathcal{T}_R|}{2}\rfloor$. We will construct from V_R and $\mathcal{PC}_L$ a (n,s,t)-constrained path cover of G of size $\pi(G_L,\mathcal{T}_L)-|V_R\setminus\mathcal{T}_R|$ such that $s=0$ if $s_L=|\mathcal{T}_R|=0$; and

$s > 0$ otherwise. The following procedure will construct such a constrained path cover:
1. partition $S_{G_L}(\mathcal{PC}_L)$ into two sets S_L^α and S_L^β such that $|S_L^\alpha| = |\mathcal{T}_R|$ and $|S_L^\beta| = s_L - |\mathcal{T}_R|$;
2. $\mathcal{PC}_{t_1} = S_L^\alpha \rhd_{|\mathcal{T}_R|} \mathcal{T}_R$;
3. partition $V_R \setminus \mathcal{T}_R$ into two sets V_R^β and V_R^γ such that $|V_R^\beta| = \lfloor \frac{s_L - |\mathcal{T}_R|}{2} \rfloor$ and $|V_R^\gamma| = |V_R \setminus \mathcal{T}_R| - \lfloor \frac{s_L - |\mathcal{T}_R|}{2} \rfloor$;
4. $\mathcal{PC}_{t_2} = V_R^\beta \ltimes_{|V_R^\beta|} S_L^\beta$;
5. let P_s be the semi-terminal path in $\mathcal{PC}_{t_2}$ if $s_L - |\mathcal{T}_R|$ is odd, and let $P_s = \emptyset$ otherwise; let $\mathcal{PC}_{t_2} = \mathcal{PC}_{t_2} \setminus \{P_s\}$;
6. $\mathcal{P}_N = N_{G_L}(\mathcal{PC}_L) \bowtie_{|V_R^\gamma|} V_R^\gamma$, where $|\mathcal{P}_N| = n_L - |V_R \setminus \mathcal{T}_R| + \lfloor \frac{s_L - |\mathcal{T}_R|}{2} \rfloor \geqslant 1$ and every path in $\mathcal{P}_N$ is a non-terminal path of G with both end vertices in V_L;
7. $\mathcal{PC}_n = \mathcal{P}_N$; $\mathcal{PC}_s = \{P_s\}$; $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup T_{G_L}(\mathcal{PC}_L)$;
8. **if** $s_L - |\mathcal{T}_R|$ is even and $|\mathcal{T}_R| \neq 0$, **then**
9. let $P_t = v_t \circ P_{\widehat{s}}$ be a path in $\mathcal{PC}_{t_1}$, where $v_t \in \mathcal{T}_R$ and $P_{\widehat{s}} \in S_L^\alpha$;
10. let P_n be a path in $\mathcal{PC}_n$;
11. $\mathcal{PC}_n = \mathcal{PC}_n \setminus \{P_n\}$; $\mathcal{PC}_s = \{P_{\widehat{s}}, v_t \circ P_n\}$; $\mathcal{PC}_t = \mathcal{PC}_t \setminus \{P_t\}$, and **goto** line 16;
12. **if** $s_L - |\mathcal{T}_R|$ is even and $s_L \neq 0$, **then** /* $|\mathcal{T}_R| = 0$ */
13. let $P_t = P_{s_1} \circ v_f \circ P_{s_2}$ be a path in $\mathcal{PC}_{t_2}$, where $P_{s_1}, P_{s_2} \in S_L^\beta$, and $v_f \in V_R$;
14. let P_n be a path in $\mathcal{PC}_n$;
15. $\mathcal{PC}_n = \mathcal{PC}_n \setminus \{P_n\}$; $\mathcal{PC}_s = \{P_{s_1} \circ v_f \circ P_n, P_{s_2}\}$; $\mathcal{PC}_t = \mathcal{PC}_t \setminus \{P_t\}$;
16. $\mathcal{PC} = \mathcal{PC}_n \cup \mathcal{PC}_s \cup \mathcal{PC}_t$.

Following the above procedure, we obtain a constrained path cover $\mathcal{PC}$ of G of size $\pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$ satisfying that it contains at least one semi-terminal path except the condition of $s_L = |\mathcal{T}_R| = 0$. Hence, $\pi(G, \mathcal{T}) \leqslant \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. By Lemma 3.5 and Statement (2) of Lemma 3.7, $\pi(G, \mathcal{T}) \geqslant \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. Hence, $\pi(G, \mathcal{T}) = \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|$. □

By the above lemma, we can construct a minimum constrained path cover of G with at least one semi-terminal path except that $2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R| < 2n_L + s_L$ and $s_L = |\mathcal{T}_R| = 0$ hold together. Next, we will show that there exists no minimum constrained path cover of G with semi-terminal path if $2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R| < 2n_L + s_L$ and $s_L = |\mathcal{T}_R| = 0$.

Lemma 3.10. *Assume that $s_L = |\mathcal{T}_R| = 0$ and $|V_R| < n_L$. Then, there exists no minimum constrained path cover of G with semi-terminal path.*

Proof. By Assumption 1, $|\mathcal{T}| = |\mathcal{T}_L| > 0$. Assume by contradiction that there exists a minimum and $(\widehat{n}, \widehat{s}, \widehat{t})$-constrained path cover $\widehat{\mathcal{PC}}$ of G with $\widehat{s} \neq 0$. Then, $V_L(\widehat{\mathcal{PC}})$ and $V_R(\widehat{\mathcal{PC}})$ are $(\widehat{n}_L, \widehat{s}_L, \widehat{t}_L)$-constrained and $(\widehat{n}_R, 0, 0)$-constrained path covers of G_L and G_R, respectively. Clearly, $\widehat{n}_R \leqslant |V_R|$. Since $\widehat{s} \neq 0$, we can see that $\widehat{s}_L \neq 0$. By Assumption 1, $V_L(\widehat{\mathcal{PC}})$ is not a minimum constrained path cover of G_L. Hence, we have that

$$\widehat{n}_L + \widehat{s}_L + \widehat{t}_L > n_L + t_L. \tag{12}$$

By Lemma 3.9, $|\widehat{\mathcal{PC}}| = \pi(G_L, \mathcal{T}_L) - |V_R| = n_L + s_L - |V_R|$. By Proposition 2.4, $|\widehat{\mathcal{PC}}| \geqslant |V_L(\widehat{\mathcal{PC}})| + |V_R(\widehat{\mathcal{PC}})| - f_{G_R}(V_R(\widehat{\mathcal{PC}})) = |V_L(\widehat{\mathcal{PC}})| + |V_R(\widehat{\mathcal{PC}})| - 2\widehat{n}_R = \widehat{n}_L + \widehat{s}_L + \widehat{t}_L - \widehat{n}_R$. By Eq. (12), we get that $n_L + s_L - |V_R| = |\widehat{\mathcal{PC}}| \geqslant \widehat{n}_L + \widehat{s}_L + \widehat{t}_L - \widehat{n}_R > n_L + t_L - \widehat{n}_R$. Thus, $|V_R| < \widehat{n}_R$ and a contradiction occurs. Therefore, there exists no minimum constrained path cover of G with semi-terminal path. □

Next, we will prove that under the condition of Assumption 2, $\pi(G, \mathcal{T}) = \lceil \frac{|\mathcal{T}|}{2} \rceil$ if $2n_L + s_L \leqslant 2n_R + s_R \leqslant 2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L|$ or $2n_R + s_R \leqslant 2n_L + s_L \leqslant 2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R|$.

Lemma 3.11. *Suppose that Assumption 2 holds. If $2n_L + s_L = 2n_R + s_R$, then $\pi(G, \mathcal{T}) = \frac{|\mathcal{T}|}{2}$.*

Proof. By Proposition 2.2, $|\mathcal{T}_L| = s_L + 2t_L$ and $|\mathcal{T}_R| = s_R + 2t_R$. By assumption of the lemma, $s_L + s_R = 2(n_R - n_L + s_R)$. Thus, $|\mathcal{T}| = |\mathcal{T}_L| + |\mathcal{T}_R| = s_L + s_R + 2(t_L + t_R) = 2(n_R - n_L + s_R + t_L + t_R)$ and hence, $|\mathcal{T}|$ is even. By Corollary 3.4, $2n_R + s_R \leqslant 2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R|$. Clearly, if $n_R = 0$ and $s_R = 0$, then $n_L = s_L = 0$ and hence, $T_{G_L}(\mathcal{PC}_L) \cup T_{G_R}(\mathcal{PC}_R)$ forms a $(0, 0, \frac{|\mathcal{T}|}{2})$-constrained path cover of G. In the following, assume that $n_R \neq 0$ or $s_R \neq 0$. We will construct a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$ from $\mathcal{PC}_L$ and $\mathcal{PC}_R$ in the following. Consider the following cases:

Case 1: $s_L = s_R$. In this case, $n_L = n_R$. There are two subcases:

Case 1.1: $s_R = 0$. By assumption, $n_R \neq 0$. Suppose that $t_R = 0$; i.e., $\mathcal{T}_R = \emptyset$. By Assumption 2, $n_L \neq |V_R|$ and hence $n_R \neq |V_R|$. Hence, we can easily obtain a $(n_R + 1, 0, 0)$-constrained path cover $\widehat{\mathcal{PC}}_R$ of G_R consisting of $n_R + 1$ non-terminal paths. Let $\mathcal{P}_N = \widehat{\mathcal{PC}}_R \bowtie_{n_L} N_{G_L}(\mathcal{PC}_L)$. Then, $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with both end vertices in V_R. Let $P_t = v_t \circ P_t'$ be a terminal path in $T_{G_L}(\mathcal{PC}_L)$, where $v_t \in \mathcal{T}_L$.

By Assumption 1, $|\mathcal{T}| = |\mathcal{T}_L| > 0$ and hence P_t exists. Let $\mathcal{PC} = T_{G_L}(\mathcal{PC}_L) \setminus \{P_t\} \cup \{v_t \circ \widehat{P} \circ P'_t\}$. Then, $\mathcal{PC}$ is a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$. On the other hand, suppose that $t_R \neq 0$. Let $\mathcal{P}_N = N_{G_R}(\mathcal{PC}_R)(\bowtie_{n_L-1}, \rhd_1)N_{G_L}(\mathcal{PC}_L)$. Then, $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with one end vertex in V_L and the other in V_R. By Assumption 1, $|\mathcal{T}_L| \geqslant |\mathcal{T}_R|$ and hence $t_L \geqslant t_R \geqslant 1$. Hence, there exist two paths $P_l = v_l \circ P'_l$ and $P_r = v_r \circ P'_r$ in $T_{G_L}(\mathcal{PC}_L)$ and $T_{G_R}(\mathcal{PC}_R)$, respectively, where $v_l \in \mathcal{T}_L$ and $v_r \in \mathcal{T}_R$. Let $\mathcal{PC} = (T_{G_L}(\mathcal{PC}_L) \setminus \{P_l\}) \cup (T_{G_R}(\mathcal{PC}_R) \setminus \{P_r\}) \cup \{P'_l \circ P'_r, v_l \circ \widehat{P} \circ v_r\}$. Then, $\mathcal{PC}$ is a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$.

Case 1.2: $s_R \neq 0$. Let $\mathcal{P}_T = S_{G_L}(\mathcal{PC}_L) \rhd_{s_R-1} S_{G_R}(\mathcal{PC}_R)$. Let P_l and P_r be two semi-terminal paths in $\mathcal{P}_T$, where $P_l \in S_{G_L}(\mathcal{PC}_L)$ and $P_r \in S_{G_R}(\mathcal{PC}_R)$, and let $\mathcal{P}_T = \mathcal{P}_T \setminus \{P_l, P_r\}$. Let $\mathcal{P}_N = N_{G_R}(\mathcal{PC}_R)(\bowtie_{n_L-1}, \rhd_1)N_{G_L}(\mathcal{PC}_L)$. Then, $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with one end vertex in V_L and the other in V_R. Let $\mathcal{PC} = \mathcal{P}_T \cup \{P_l \circ \widehat{P} \circ P_r\} \cup T_{G_L}(\mathcal{PC}_L) \cup T_{G_R}(\mathcal{PC}_R)$. Then, $\mathcal{PC}$ is a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$.

Case 2: $s_L > s_R$. In this case, $s_L - s_R = 2(n_R - n_L) > 0$ and hence, $n_R - n_L > 0$. We first partition $S_{G_L}(\mathcal{PC}_L)$ into two sets S_L^α and S_L^β such that $|S_L^\alpha| = s_R$ and $|S_L^\beta| = s_L - s_R$. Then we perform the following procedure: (1) $\mathcal{PC}_{t_1} = S_L^\alpha \rhd_{s_R} S_{G_R}(\mathcal{PC}_R)$; (2) $\mathcal{P}_N = N_{G_R}(\mathcal{PC}_R) \bowtie_{n_L} N_{G_L}(\mathcal{PC}_L)$, where $|\mathcal{P}_N| = n_R - n_L$; (3) $\mathcal{PC}_{t_2} = \mathcal{P}_N \ltimes_{n_R-n_L} S_L^\beta$, and (4) $\mathcal{PC} = \mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup T_{G_L}(\mathcal{PC}_L) \cup T_{G_R}(\mathcal{PC}_R)$. Then, $\mathcal{PC}$ is a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$.

Case 3: $s_L < s_R$. By symmetry, we can obtain a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$ by arguments similar to those used in proving Case 2. □

Lemma 3.12. *Suppose that Assumption* 2 *holds. Assume that* $n_R = n_L + \lceil \frac{s_L - s_R}{2} \rceil$ *if* $s_L \geqslant s_R$*; and* $n_L = n_R + \lceil \frac{s_R - s_L}{2} \rceil$ *otherwise. Then,* $\pi(G, \mathcal{T}) = \lceil \frac{|\mathcal{T}|}{2} \rceil$.

Proof. First consider the case of $s_L \geqslant s_R$. Then, $n_R = n_L + \lceil \frac{s_L - s_R}{2} \rceil$. If $s_L - s_R$ is even, then $2n_L + s_L = 2n_R + s_R$ and hence, we can construct a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$ from $\mathcal{PC}_L$ and $\mathcal{PC}_R$ by Lemma 3.11. Suppose that $s_L - s_R$ is odd. Let P_s be a path in $S_{G_L}(\mathcal{PC}_L)$, $\widetilde{\mathcal{PC}}_L = \mathcal{PC}_L \setminus \{P_s\}$, and let $\widetilde{G} = G \setminus \{P_s\}$. By using similar arguments for proving Lemma 3.11, we can obtain a constrained path cover $\widetilde{\mathcal{PC}}$ of $\widetilde{G}$ of size $\lfloor \frac{|\mathcal{T}|}{2} \rfloor$ from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$. Let $\mathcal{PC} = \widetilde{\mathcal{PC}} \cup \{P_s\}$. Then, $\mathcal{PC}$ forms a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$.

We next consider the case of $s_L < s_R$. Then, $n_L = n_R + \lceil \frac{s_R - s_L}{2} \rceil$. If $s_R - s_L$ is even, then $2n_L + s_L = 2n_R + s_R$ and hence, we can construct a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$ from $\mathcal{PC}_L$ and $\mathcal{PC}_R$ by Lemma 3.11. Suppose that $s_R - s_L$ is odd. Let $s'_R = s_R - 1$. Then, $s'_R - s_L$ is even. The case of $s'_R > s_L$ can be proved by using similar arguments for proving Lemma 3.11. Assume that $s'_R = s_L$ below. Then, $s_L = s_R - 1$ and $n_L = n_R + 1$. We then construct a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ from $\mathcal{PC}_L$ and $\mathcal{PC}_R$ as follows: (1) let P_s be a path in $S_{G_R}(\mathcal{PC}_R)$; (2) $\widetilde{S}_R = S_{G_R}(\mathcal{PC}_R) \setminus \{P_s\}$; (3) $\mathcal{PC}_{t_1} = \widetilde{S}_R \rhd_{s_L} S_{G_L}(\mathcal{PC}_L)$; (4) $\mathcal{P}_N = N_{G_L}(\mathcal{PC}_L) \bowtie_{n_R} N_{G_R}(\mathcal{PC}_R)$, where $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with both end vertices in V_L; (5) $P_s = P_s \circ \widehat{P}$; (6) $\mathcal{PC} = \mathcal{PC}_{t_1} \cup T_{G_L}(\mathcal{PC}_L) \cup T_{G_R}(\mathcal{PC}_R) \cup \{P_s\}$. Then, $\mathcal{PC}$ is a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$. □

Lemma 3.13. *Suppose that Assumption* 2 *holds. If* $2n_L + s_L < 2n_R + s_R \leqslant 2|V_L \setminus \mathcal{T}_L| + |\mathcal{T}_L|$*, then* $\pi(G, \mathcal{T}) = \lceil \frac{|\mathcal{T}|}{2} \rceil$.

Proof. By Proposition 2.2, $|\mathcal{T}_L| = s_L + 2t_L$. Hence, $2n_L + s_L < 2n_R + s_R \leqslant 2|V_L \setminus \mathcal{T}_L| + s_L + 2t_L$. Then, we have that

$$n_L + \lceil \tfrac{s_L - s_R}{2} \rceil \leqslant n_R \leqslant |V_L \setminus \mathcal{T}_L| + \lceil \tfrac{s_L - s_R}{2} \rceil + t_L, \quad if s_L \geqslant s_R; \quad (13)$$

$$n_L < n_R + \lceil \tfrac{s_R - s_L}{2} \rceil \leqslant |V_L \setminus \mathcal{T}_L| + t_L, \quad if s_L < s_R. \quad (14)$$

We prove the lemma by showing how to construct a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ from $\widehat{\mathcal{PC}}_L$ and $\mathcal{PC}_R$, where $\widehat{\mathcal{PC}}_L$ may not be a minimum constrained path cover of G_L. Consider the following cases:

Case 1: $s_L \geqslant s_R$. Suppose that $n_R = n_L + \lceil \frac{s_L - s_R}{2} \rceil$. By Lemma 3.12, we can obtain a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$. In the following, assume that $n_R > n_L + \lceil \frac{s_L - s_R}{2} \rceil$. By Eq. (13), $n_L < n_R - \lceil \frac{s_L - s_R}{2} \rceil \leqslant |V_L \setminus \mathcal{T}_L| + t_L$. Let $\mathcal{NS} = N_{G_L}(\mathcal{PC}_L) \cup S_{G_L}(\mathcal{PC}_L)$. Recall that $F(\mathcal{NS})$ is the set of free vertices in $\mathcal{NS}$. There are two subcases:

Case 1.1: $|F(\mathcal{NS})| \geqslant n_R - \lceil \frac{s_L - s_R}{2} \rceil$. In this subcase, $|F(\mathcal{NS})| \geqslant n_R - \lceil \frac{s_L - s_R}{2} \rceil > n_L$. By Proposition 2.3, we can obtain from $\mathcal{NS}$ a set

$\widehat{\mathcal{NS}}$ of vertex-disjoint paths with $\widehat{n}_L$ non-terminal paths and s_L semi-terminal paths such that $\widehat{n}_L = n_R - \lceil\frac{s_L-s_R}{2}\rceil$ if $s_L \neq 0$; and $\widehat{n}_L = n_R - 1$ otherwise. Note that if $s_L = 0$, then $s_R = 0$ since $s_L \geqslant s_R$. Let $\widehat{\mathcal{PC}}_L = \widehat{\mathcal{NS}} \cup T_{G_L}(\mathcal{PC}_L)$. Then, $\widehat{\mathcal{PC}}_L$ is a $(\widehat{n}_L, s_L, t_L)$-constrained path cover of G_L. Let $\widehat{\mathcal{N}}_L = N_{G_L}(\widehat{\mathcal{PC}}_L)$ and $\widehat{\mathcal{S}}_L = S_{G_L}(\widehat{\mathcal{PC}}_L)$. We can obtain a constrained path cover $\mathcal{PC}$ of G of size $\lceil\frac{|\mathcal{T}|}{2}\rceil$ from $\widehat{\mathcal{PC}}_L$ and $\mathcal{PC}_R$ through the following procedure:

1. **if** $s_L = 0$, **then** /* $\widehat{n}_L = n_R - 1$ */
2. $\mathcal{P}_N = N_{G_R}(\mathcal{PC}_R) \bowtie_{n_R-1} \widehat{\mathcal{N}}_L$, where $|\widehat{\mathcal{N}}_L| = n_R - 1$ and $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with both end vertices in V_R;
3. let $P_t = v_t \circ P'_t$ be a terminal path in $T_{G_L}(\widehat{\mathcal{PC}}_L)$, where $v_t \in \mathcal{T}_L$ and P_t exists since $s_L = 0$, $|\mathcal{T}_L| \geqslant |\mathcal{T}_R|$, and $|\mathcal{T}_L| + |\mathcal{T}_R| \geqslant 1$ by Assumption 1;
4. $\mathcal{PC}_{t_1} = T_{G_L}(\widehat{\mathcal{PC}}_L)$ and $\mathcal{PC}_{t_1} = \mathcal{PC}_{t_1} \setminus \{P_t\} \cup \{v_t \circ \widehat{P} \circ P'_t\}$;
5. $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup T_{G_R}(\mathcal{PC}_R)$ and $\mathcal{PC} = \mathcal{PC}_t$;
6. **else** /* $\widehat{n}_L = n_R - \lceil\frac{s_L-s_R}{2}\rceil$ */
7. partition $\widehat{\mathcal{S}}_L$ into two sets S_L^α and S_L^β such that $|S_L^\alpha| = s_R$ and $|S_L^\beta| = s_L - s_R$;
8. $\mathcal{PC}_{t_1} = S_L^\alpha \rhd_{s_R} S_{G_R}(\mathcal{PC}_R)$;
9. $\mathcal{P}_N = N_{G_R}(\mathcal{PC}_R) \ltimes_{\lfloor\frac{s_L-s_R}{2}\rfloor} S_L^\beta$;
10. **if** $s_L - s_R$ is odd, **then** let P_s and P_n be the semi-terminal path and non-terminal path in $\mathcal{P}_N$, respectively; let $P = P_s \circ P_n$, $\mathcal{PC}_s = \{P\}$, and let $\mathcal{P}_N = \mathcal{P}_N \setminus \{P_s, P_n\}$;
11. partition $\mathcal{P}_N$ into two sets $\mathcal{PC}_{t_2}$ and $\mathcal{N}_R$ such that $\mathcal{PC}_{t_2}$ contains all terminal paths in $\mathcal{P}_N$;
12. $\mathcal{P} = \mathcal{N}_R(\bowtie_{|\widehat{\mathcal{N}}_L|-1}, \rhd_1)\widehat{\mathcal{N}}_L$, where $|\mathcal{N}_R| = |\widehat{\mathcal{N}}_L| = n_R - \lceil\frac{s_L-s_R}{2}\rceil \geqslant 1$ and $\mathcal{P}$ contains only a non-terminal path $\widehat{P}$ of G with one end vertex in V_L and the other in V_R;
13. let $P = P_s \circ P'$ be a path in $\mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup \mathcal{PC}_s$ such that P_s is a semi-terminal path in $\widehat{\mathcal{S}}_L$, where P exists since $s_L > 0$;
14. let $\mathcal{Y}$ be the set containing P and let $\mathcal{Y} = \mathcal{Y} \setminus \{P\} \cup \{P_s \circ \widehat{P} \circ P'\}$;
15. $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup T_{G_L}(\widehat{\mathcal{PC}}_L) \cup T_{G_R}(\mathcal{PC}_R)$ and $\mathcal{PC} = \mathcal{PC}_t \cup \mathcal{PC}_s$.

Following the above procedure, we can construct a constrained path cover $\mathcal{PC}$ of G of size $\lceil\frac{|\mathcal{T}|}{2}\rceil$.

Case 1.2: $|F(\mathcal{NS})| < n_R - \lceil\frac{s_L-s_R}{2}\rceil$. By definition, $V_L \setminus \mathcal{T}_L = F(\mathcal{NS}) \cup F(T_{G_L}(\mathcal{PC}_L))$. Let $\widehat{\mathcal{N}}_L^v$ and $\widehat{\mathcal{S}}_L^v$ denote the sets of free vertices and terminals in $\mathcal{NS}$, respectively. Then, $|\widehat{\mathcal{N}}_L^v| = |F(\mathcal{NS})|$ and $|\widehat{\mathcal{S}}_L^v| = s_L$. Let $\mathcal{T}_{G_L}(\mathcal{PC}_L) = \{P_1, P_2, \cdots, P_{t_L}\}$, where $P_i = v_i \circ P'_i$ and $v_i \in \mathcal{T}_L$ for $t_L \geqslant i \geqslant 1$. By definition, every path P_i in $\mathcal{T}_{G_L}(\mathcal{PC}_L)$ can be split into two semi-terminal paths v_i and P'_i of G_L. There are two subcases:

Case 1.2.1: $n_R - \lceil\frac{s_L-s_R}{2}\rceil \leqslant |F(\mathcal{NS})| + t_L$. In this subcase, $|F(\mathcal{NS})| + t_L \geqslant n_R - \lceil\frac{s_L-s_R}{2}\rceil > |F(\mathcal{NS})|$. Thus, there exists a number k such that $n_R - \lceil\frac{s_L-s_R}{2}\rceil = |F(\mathcal{NS})| + k$ and $t_L \geqslant k \geqslant 1$. Let $\mathcal{P}_T = \{P_1, P_2, \cdots, P_k\}$ be a subset of $T_{G_L}(\mathcal{PC}_L)$. For every path $P_i = v_i \circ P'_i$ in $\mathcal{P}_T$, we first split P_i into two semi-terminal paths v_i and P'_i. Then, we get a set $\mathcal{P}_S$ of semi-terminal paths $\{v_1, v_2, \cdots, v_k, P'_1, P'_2, \cdots, P'_k\}$. Let $\widetilde{\mathcal{P}}_T = \mathcal{T}_{G_L}(\mathcal{PC}_L) \setminus \mathcal{P}_T$, $\widetilde{\mathcal{P}}_S = \widehat{\mathcal{S}}_L^v \cup \mathcal{P}_S$, $\widetilde{\mathcal{P}}_N = \widehat{\mathcal{N}}_L^v$, and let $\widetilde{\mathcal{PC}}_L = \widetilde{\mathcal{P}}_N \cup \widetilde{\mathcal{P}}_S \cup \widetilde{\mathcal{P}}_T$. Let $\widetilde{n}_L = |F(\mathcal{NS})|$, $\widetilde{s}_L = s_L + 2k$, and let $\widetilde{t}_L = t_L - k$. Then, $\widetilde{\mathcal{PC}}_L$ is a $(\widetilde{n}_L, \widetilde{s}_L, \widetilde{t}_L)$-constrained path cover of G_L satisfying that $n_R - \lceil\frac{s_L-s_R}{2}\rceil = \widetilde{n}_L + k$ and hence $n_R = \widetilde{n}_L + \lceil\frac{\widetilde{s}_L-s_R}{2}\rceil$. Since $s_L \geqslant s_R$ and $k \geqslant 1$, we get that $\widetilde{s}_L = s_L + 2k > s_R$. By arguments similar to those used in proving Lemma 3.12, we can construct a constrained path cover of G of size $\lceil\frac{|\mathcal{T}|}{2}\rceil$ from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$.

Case 1.2.2: $n_R - \lceil\frac{s_L-s_R}{2}\rceil > |F(\mathcal{NS})| + t_L$. In this subcase, we first find a minimum k, where $1 \leqslant k \leqslant t_L$, such that $|F(\cup_{i=1}^k P_i)| + |F(\mathcal{NS})| \geqslant n_R - \lceil\frac{(s_L+2k)-s_R}{2}\rceil$. Note that such a number k exists since $|F(\mathcal{NS})| + t_L < n_R - \lceil\frac{s_L-s_R}{2}\rceil \leqslant |V_L \setminus \mathcal{T}_L| + t_L = |F(\mathcal{NS})| + |F(T_{G_L}(\widehat{\mathcal{PC}}_L))| + t_L$. Let $\mathcal{P}_T = \{P_1, P_2, \cdots, P_k\}$ be a subset of $T_{G_L}(\mathcal{PC}_L)$. For every path $P_i = v_i \circ P'_i$ in $\mathcal{P}_T$, we first split P_i into two semi-terminal paths v_i and P'_i. Then, we get a set $\mathcal{P}_S$ of semi-terminal paths $\{v_1, v_2, \cdots, v_k, P'_1, P'_2, \cdots, P'_k\}$. Let $\widetilde{\mathcal{P}}_T = \mathcal{T}_{G_L}(\mathcal{PC}_L) \setminus \mathcal{P}_T$, $\widetilde{\mathcal{P}}_S = \widehat{\mathcal{S}}_L^v \cup \mathcal{P}_S$, $\widetilde{\mathcal{P}}_N = \widehat{\mathcal{N}}_L^v$, and let $\widetilde{\mathcal{PC}}_L = \widetilde{\mathcal{P}}_N \cup \widetilde{\mathcal{P}}_S \cup \widetilde{\mathcal{P}}_T$. Let $\widetilde{\mathcal{NS}} = \widetilde{\mathcal{P}}_N \cup \widetilde{\mathcal{P}}_S$, $\widetilde{n}_L = |F(\mathcal{NS})|$, $\widetilde{s}_L = s_L + 2k$, and let $\widetilde{t}_L = t_L - k$. Then, $|F(\widetilde{\mathcal{NS}})| = |F(\cup_{i=1}^k P_i)| + |F(\mathcal{NS})| \geqslant n_R - \lceil\frac{\widetilde{s}_L-s_R}{2}\rceil$. Since $t_L \geqslant k$ and $n_R - \lceil\frac{s_L-s_R}{2}\rceil > |F(\widetilde{\mathcal{NS}})| + t_L = \widetilde{n}_L + t_L$, we have that $n_R - \lceil\frac{s_L+2k-s_R}{2}\rceil > \widetilde{n}_L$. Hence, $\widetilde{\mathcal{PC}}_L$ is a $(\widetilde{n}_L, \widetilde{s}_L, \widetilde{t}_L)$-constrained path cover of G_L satisfying that $|F(\widetilde{\mathcal{NS}})| \geqslant n_R - \lceil\frac{\widetilde{s}_L-s_R}{2}\rceil > \widetilde{n}_L$ and $\widetilde{s}_L > s_R$. Then, we can construct a constrained path cover of G of size $\lceil\frac{|\mathcal{T}|}{2}\rceil$ from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$ by using arguments similar to those used in proving Case 1.1.

Case 2: $s_L < s_R$. By Eq. (14), $n_L < n_R + \lceil\frac{s_R-s_L}{2}\rceil \leqslant |V_L \setminus \mathcal{T}_L| + t_L$. Let $\mathcal{NS} = N_{G_L}(\mathcal{PC}_L) \cup S_{G_L}(\mathcal{PC}_L)$. There are two subcases:

Case 2.1: $|F(\mathcal{NS})| \geqslant n_R + \lceil\frac{s_R-s_L}{2}\rceil$. In this subcase, $n_L < n_R + \lceil\frac{s_R-s_L}{2}\rceil \leqslant |F(\mathcal{NS})|$. By Proposition 2.3, we can obtain from $\mathcal{NS}$ a set

$\widehat{\mathcal{NS}}$ of vertex-disjoint paths with $\widehat{n}_L$ non-terminal paths and s_L semi-terminal paths such that $\widehat{n}_L = n_R + \lceil \frac{s_R - s_L}{2} \rceil$. Let $\widehat{\mathcal{N}}_L$ and $\widehat{\mathcal{S}}_L$ be the sets in $\widehat{\mathcal{NS}}$ containing all non-terminal and semi-terminal paths, respectively. We then construct a constrained path cover $\mathcal{PC}$ of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ through the following procedure:
1. partition $S_{G_R}(\mathcal{PC}_R)$ into two sets S_R^α and S_R^β such that $|S_R^\alpha| = s_L$ and $|S_R^\beta| = s_R - s_L$;
2. partition $\widehat{\mathcal{N}}_L$ into two sets N_L^β and N_L^γ such that $|N_L^\beta| = \lceil \frac{s_R - s_L}{2} \rceil$ and $|N_L^\gamma| = n_R$;
3. $\mathcal{PC}_{t_1} = \widehat{\mathcal{S}}_L \rhd_{s_L} S_R^\alpha$;
4. $\mathcal{PC}_{t_2} = N_L^\beta \ltimes_{\lfloor \frac{s_R - s_L}{2} \rfloor} S_R^\beta$;
5. **if** $s_R - s_L$ is odd, **then** let P_s and P_n be the semi-terminal path and non-terminal path in $\mathcal{PC}_{t_2}$, respectively; let $P = P_s \circ P_n$, $\mathcal{PC}_s = \{P\}$, and let $\mathcal{PC}_{t_2} = \mathcal{PC}_{t_2} \setminus \{P_s, P_n\}$;
6. $\mathcal{P}_N = N_L^\gamma(\bowtie_{n_R-1}, \rhd_1) N_{G_R}(\mathcal{PC}_R)$, where $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with one end vertex in V_L and the other in V_R if $n_R \neq 0$, and $\mathcal{P}_N = \emptyset$ otherwise;
7. **if** $\mathcal{P}_N \neq \emptyset$, **then**
8. let $P_t = P_s \circ P_t'$ be a path in $\mathcal{PC}_s \cup \mathcal{PC}_{t_2}$ such that $P_s \in S_R^\beta$, where P_t exists since $|S_R^\beta| = s_R - s_L > 0$;
9. let $\mathcal{Y}$ be the set containing P_t and let $\mathcal{Y} = \mathcal{Y} \setminus \{P_t\} \cup \{P_s \circ \widehat{P} \circ P_t'\}$;
10. $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup T_{G_L}(\mathcal{PC}_L) \cup T_{G_R}(\mathcal{PC}_R)$ and $\mathcal{PC} = \mathcal{PC}_t \cup \mathcal{PC}_s$.

Following the above procedure, we can obtain a constrained path cover $\mathcal{PC}$ of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$.

Case 2.2: $|F(\mathcal{NS})| < n_R + \lceil \frac{s_R - s_L}{2} \rceil$. By definition, $V_L \setminus \mathcal{T}_L = F(\mathcal{NS}) \cup F(T_{G_L}(\mathcal{PC}_L))$. Let $\widehat{\mathcal{N}}_L^v$ and $\widehat{\mathcal{S}}_L^v$ denote the sets of free vertices and terminals in $\mathcal{NS}$, respectively. Then, $|\widehat{\mathcal{N}}_L^v| = |F(\mathcal{NS})|$ and $|\widehat{\mathcal{S}}_L^v| = s_L$. Let $\mathcal{T}_{G_L}(\mathcal{PC}_L) = \{P_1, P_2, \cdots, P_{t_L}\}$. By definition, every path in $\mathcal{T}_{G_L}(\mathcal{PC}_L)$ can be split into two semi-terminal paths. There are two subcases:

Case 2.2.1: $n_R + \lceil \frac{s_R - s_L}{2} \rceil \leqslant |F(\mathcal{NS})| + t_L$. By similar arguments for proving Case 1.2.1, we can obtain a $(\widetilde{n}_L, \widetilde{s}_L, \widetilde{t}_L)$-constrained path cover $\widetilde{\mathcal{PC}}_L$ of G_L satisfying that $n_R + \lceil \frac{s_R - \widetilde{s}_L}{2} \rceil = \widetilde{n}_L$. If $s_R > \widetilde{s}_L$, then we can construct from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$ a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ by using arguments similar to those used in proving Lemma 3.12. If $s_R \leqslant \widetilde{s}_L$ and $\widetilde{s}_L - s_R$ is even, then we can construct from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$ a constrained path cover of G of size $\frac{|\mathcal{T}|}{2}$ by using arguments similar to those used in proving Lemma 3.11. Suppose that $s_R \leqslant \widetilde{s}_L$ and $\widetilde{s}_L - s_R$ is odd. Let $s_L' = \widetilde{s}_L - 1$. Then, $s_L' - s_R$ is even and $n_R + \lceil \frac{s_R - (s_L'+1)}{2} \rceil = n_R + \frac{s_R - s_L'}{2} = \widetilde{n}_L$. Thus, $n_R = \widetilde{n}_L + \frac{s_L' - s_R}{2}$. By similar arguments for proving Lemma 3.12, we can construct from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$ a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$.

Case 2.2.2: $n_R + \lceil \frac{s_R - s_L}{2} \rceil > |F(\mathcal{NS})| + t_L$. By similar arguments for proving Case 1.2.2, we can obtain a $(\widetilde{n}_L, \widetilde{s}_L, \widetilde{t}_L)$-constrained path cover $\widetilde{\mathcal{PC}}_L$ of G_L satisfying that $F(N_{G_L}(\widetilde{\mathcal{PC}}_L) \cup S_{G_L}(\widetilde{\mathcal{PC}}_L)) \geqslant n_R + \lceil \frac{s_R - \widetilde{s}_L}{2} \rceil > \widetilde{n}_L$, where $\widetilde{t}_L < t_L$. Depending on whether $\widetilde{s}_L \geqslant s_R$ or not, we can construct a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ from $\widetilde{\mathcal{PC}}_L$ and $\mathcal{PC}_R$ by using arguments similar to those used in proving Case 1.1 or Case 2.1. □

Lemma 3.14. *Suppose that Assumption* 2 *holds. If* $2n_R + s_R < 2n_L + s_L \leqslant 2|V_R \setminus \mathcal{T}_R| + |\mathcal{T}_R|$, *then* $\pi(G, \mathcal{T}) = \lceil \frac{|\mathcal{T}|}{2} \rceil$.

Proof. This lemma can be proved by similar arguments for proving Lemma 3.13 except one case. By Proposition 2.2, $|\mathcal{T}_R| = s_R + 2t_R$. Hence, $2n_R + s_R < 2n_L + s_L \leqslant 2|V_R \setminus \mathcal{T}_R| + s_R + 2t_R$. Then, we have that

$$n_R + \lceil \tfrac{s_R - s_L}{2} \rceil \leqslant n_L \leqslant |V_R \setminus \mathcal{T}_R| + \lceil \tfrac{s_R - s_L}{2} \rceil + t_R, \quad if s_R \geqslant s_L; \quad (15)$$

$$n_R < n_L + \lceil \tfrac{s_L - s_R}{2} \rceil \leqslant |V_R \setminus \mathcal{T}_R| + t_R, \quad if s_R < s_L. \quad (16)$$

By similar arguments for proving Lemma 3.13, we show that a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ can be constructed from $\widehat{\mathcal{PC}}_R$ and $\mathcal{PC}_L$, where $\widehat{\mathcal{PC}}_R$ may not be a minimum constrained path cover of G_R. Consider the following cases:
Case 1: $s_R \geqslant s_L$. Suppose that $n_L = n_R + \lceil \frac{s_R - s_L}{2} \rceil$. By Lemma 3.12, we can obtain a constrained path cover of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$. In the following, assume that $n_L > n_R + \lceil \frac{s_R - s_L}{2} \rceil$. By Eq. (15), $n_R < n_L - \lceil \frac{s_R - s_L}{2} \rceil \leqslant |V_R \setminus \mathcal{T}_R| + t_R$. Let $\mathcal{NS} = N_{G_R}(\mathcal{PC}_R) \cup S_{G_R}(\mathcal{PC}_R)$. There are two subcases:

Case 1.1: $|F(\mathcal{NS})| \geqslant n_L - \lceil \frac{s_R - s_L}{2} \rceil$. In this subcase, $|F(\mathcal{NS})| \geqslant n_L - \lceil \frac{s_R - s_L}{2} \rceil > n_R$. By Proposition 2.3, we can obtain from $\mathcal{NS}$ a set $\widehat{\mathcal{NS}}$ of vertex-disjoint paths with $\widehat{n}_R$ non-terminal paths and s_R semi-terminal paths such that $\widehat{n}_R = n_L - \lceil \frac{s_R - s_L}{2} \rceil$ if $|\mathcal{T}_R| > 0$; and $\widehat{n}_R = n_L + 1$ otherwise. Consider the case of $|\mathcal{T}_R| = 0$. Then, $s_R = s_L = 0$. By Assumption 2, $n_L \neq |V_R|$ and hence, $|V_R| > n_L > n_R$. Thus, there exists a number $\widehat{n}_R$ such that $\widehat{n}_R = n_L + 1$.

Let $\widehat{\mathcal{PC}}_R = \widehat{\mathcal{NS}} \cup T_{G_R}(\mathcal{PC}_R)$. Then, $\widehat{\mathcal{PC}}_R$ is a $(\widehat{n}_R, s_R, t_R)$-constrained path cover of G_R. Let $\widehat{\mathcal{N}}_R = N_{G_R}(\widehat{\mathcal{PC}}_R)$ and $\widehat{\mathcal{S}}_R = S_{G_R}(\widehat{\mathcal{PC}}_R)$. We can obtain a constrained path cover $\mathcal{PC}$ of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$ from $\widehat{\mathcal{PC}}_R$ and $\mathcal{PC}_L$ through the following procedure:

1. **if** $|\mathcal{T}_R| = 0$, **then** /* $\widehat{n}_R = n_L + 1$ */
2. $\mathcal{P}_N = \widehat{\mathcal{N}}_R \bowtie_{n_L} N_{G_L}(\mathcal{PC}_L)$, where $|\widehat{\mathcal{N}}_R| = n_L + 1$ and $\mathcal{P}_N$ contains only a non-terminal path $\widehat{P}$ of G with both end vertices in V_R;
3. let $P_t = v_t \circ P'_t$ be a terminal path in $T_{G_L}(\mathcal{PC}_L)$, where $v_t \in \mathcal{T}_L$ and P_t exists since $|\mathcal{T}_L| \geqslant 1$ by Assumption 1;
4. $\mathcal{PC}_{t_1} = T_{G_L}(\mathcal{PC}_L)$ and $\mathcal{PC}_{t_1} = \mathcal{PC}_{t_1} \setminus \{P_t\} \cup \{v_t \circ \widehat{P} \circ P'_t\}$;
5. $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup T_{G_R}(\widehat{\mathcal{PC}}_R)$ and $\mathcal{PC} = \mathcal{PC}_t$;
6. **else** /* $\widehat{n}_R = n_L - \lceil \frac{s_R - s_L}{2} \rceil$ */
7. partition $\widehat{\mathcal{S}}_R$ into two sets S^α_R and S^β_R such that $|S^\alpha_R| = s_L$ and $|S^\beta_R| = s_R - s_L$;
8. $\mathcal{PC}_{t_1} = S^\alpha_R \rhd_{s_L} S_{G_L}(\mathcal{PC}_L)$;
9. $\mathcal{P}_N = N_{G_L}(\mathcal{PC}_L) \ltimes_{\lfloor \frac{s_R - s_L}{2} \rfloor} S^\beta_R$;
10. **if** $s_R - s_L$ is odd, **then** let P_s and P_n be the semi-terminal path and non-terminal path in $\mathcal{P}_N$, respectively; let $P = P_s \circ P_n$, $\mathcal{PC}_s = \{P\}$, and let $\mathcal{P}_N = \mathcal{P}_N \setminus \{P_s, P_n\}$;
11. partition $\mathcal{P}_N$ into two sets $\mathcal{PC}_{t_2}$ and $\mathcal{N}_L$ such that $\mathcal{PC}_{t_2}$ contains all terminal paths in $\mathcal{P}_N$;
12. $\mathcal{P} = \mathcal{N}_L(\bowtie_{|\widehat{\mathcal{N}}_R|-1}, \rhd_1)\widehat{\mathcal{N}}_R$, where $|\mathcal{N}_L| = |\widehat{\mathcal{N}}_R| = n_L - \lceil \frac{s_R - s_L}{2} \rceil \geqslant 1$ and $\mathcal{P}$ contains only a non-terminal path $\widehat{P}$ of G with one end vertex in V_L and the other in V_R;
13. **if** $t_R > 0$, **then**
14. let $P_l = v_l \circ P'_l$ and $P_r = v_r \circ P'_r$ be the terminal paths in $T_{G_L}(\mathcal{PC}_L)$ and $T_{G_R}(\widehat{\mathcal{PC}}_R)$, respectively, where $v_l \in \mathcal{T}_L$, $v_r \in \mathcal{T}_R$, and P_l exists since $|\mathcal{T}_L| \geqslant |\mathcal{T}_R|$ by Assumption 1;
15. $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup (T_{G_L}(\mathcal{PC}_L) \setminus \{P_l\}) \cup (T_{G_R}(\widehat{\mathcal{PC}}_R) \setminus \{P_r\}) \cup \{P'_l \circ P'_r, v_l \circ \widehat{P} \circ v_r\}$;
16. **else** /* $t_R = 0$ and $s_R > 0$ */
17. let $P = P_s \circ P'$ be a path in $\mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup \mathcal{PC}_s$ such that P_s is a semi-terminal path in $\widehat{\mathcal{S}}_R$, where P exists since $s_R > 0$;
18. let $\mathcal{Y}$ be the set containing P and let $\mathcal{Y} = \mathcal{Y} \setminus \{P\} \cup \{P_s \circ \widehat{P} \circ P'\}$;
19. $\mathcal{PC}_t = \mathcal{PC}_{t_1} \cup \mathcal{PC}_{t_2} \cup T_{G_L}(\mathcal{PC}_L) \cup T_{G_R}(\widehat{\mathcal{PC}}_R)$;
20. $\mathcal{PC} = \mathcal{PC}_t \cup \mathcal{PC}_s$.

Following the above procedure, we can construct a constrained path cover $\mathcal{PC}$ of G of size $\lceil \frac{|\mathcal{T}|}{2} \rceil$.

Case 1.2: $|F(\mathcal{NS})| < n_L - \lceil \frac{s_R - s_L}{2} \rceil$. By symmetry, this subcase can be proved by using arguments similar to those used in proving Case 1.2 of Lemma 3.13.

Case 2: $s_R < s_L$. By similar arguments for proving Case 2 of Lemma 3.13, this case can be proved. □

Following Lemma 3.6, Lemmas 3.8–3.9, and Lemmas 3.11–3.14, we conclude the following theorem:

Theorem 3.15. *Let $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ be two cographs with terminal sets $\mathcal{T}_L$ and $\mathcal{T}_R$, respectively, and let $G = G_L \otimes G_R$. Assume that $|\mathcal{T}_L| \geqslant |\mathcal{T}_R|$, $|\mathcal{T}| = |\mathcal{T}_L| + |\mathcal{T}_R| \geqslant 1$, and that $\mathcal{PC}_L$ (resp. $\mathcal{PC}_R$) is a minimum and (n_L, s_L, t_L)-constrained (resp. (n_R, s_R, t_R)-constrained) path cover of G_L (resp. G_R) satisfying that there exists no minimum constrained path cover of G_L (resp. G_R) with semi-terminal path if $s_L = 0$ (resp. $s_R = 0$). Then,*

$$\pi(G, \mathcal{T}) = \begin{cases} \frac{|\mathcal{T}|}{2} + 1 & ,\textit{if } s_L = |\mathcal{T}_R| = 0 \textit{ and } n_L = |V_R|; \\ \max\{\lceil \frac{|\mathcal{T}|}{2} \rceil, \pi(G_L, \mathcal{T}_L) - |V_R \setminus \mathcal{T}_R|, \pi(G_R, \mathcal{T}_R) - |V_L \setminus \mathcal{T}_L|\} & ,\textit{otherwise.} \end{cases}$$

Based upon Lemma 3.2 and Theorem 3.15, we can construct a minimum constrained path cover of a cograph G in linear time. In the following, we will show how to find such a minimum constrained path cover of G. For a cograph G, a decomposition tree $DT(G)$ of G can be constructed in linear time [7, 10]. Note that the leaves of $DT(G)$ are the vertices of G and every internal node in $DT(G)$ is labeled by either '$\oplus$' or '$\otimes$'. In the following, assume that a decomposition tree $DT(G)$ of a cograph G and a terminal set $\mathcal{T}$ of G are given. For a node v in $DT(G)$, let $DT_v(G)$ denote the subtree of $DT(G)$ rooted at v, G_v denote the subgraph of G induced by the leaves of $DT_v(G)$, and let $\mathcal{T}_v$ denote the set of terminals in G_v. Our algorithm finds a minimum constrained path cover $\mathcal{PC}_v$ of G_v for each $v \in DT(G)$ such that there exists no minimum constrained path cover of G_v with semi-terminal path if $S_{G_v}(\mathcal{PC}_v) = \emptyset$, and is sketched as follows: The algorithm visits nodes of $DT(G)$ in a postorder sequence (i.e., bottom-up). Thus, when visiting an internal node of $DT(G)$, both its children were visited. Assume that it is about to process internal node v. Let v_L and v_R be the children of v in $DT(G)$, and let V_L and V_R be the vertex sets of G_{v_L} and G_{v_R}, respectively. Without loss of generality, assume that $|\mathcal{T}_{v_L}| \geqslant |\mathcal{T}_{v_R}|$.

Our algorithm maintains the invariant that a minimum constrained path cover $\mathcal{PC}_{v_L}$ (resp. $\mathcal{PC}_{v_R}$) of G_{v_L} (resp. G_{v_R}) is constructed and satisfies that there exists no minimum constrained path cover of G_{v_L} (resp. G_{v_R}) with semi-terminal path if $S_{G_{v_L}}(\mathcal{PC}_{v_L}) = \emptyset$ (resp. $S_{G_{v_R}}(\mathcal{PC}_{v_R}) = \emptyset$). Let $\mathcal{PC}_{v_L}$ and $\mathcal{PC}_{v_R}$ be (n_L, s_L, t_L)-constrained and (n_R, s_R, t_R)-constrained path covers of G_{v_L} and G_{v_R}, respectively. Suppose that v is labeled by '$\oplus$'. Then, $\mathcal{PC}_{v_L} \cup \mathcal{PC}_{v_R}$ forms a minimum and $(n_L + n_R, s_L + s_R, t_L + t_R)$-constrained path cover of G_v satisfying that there exists no minimum constrained path cover of G_v with semi-terminal path if $s_L + s_R = 0$. In the following, assume that v is labeled by '$\otimes$', i.e., $G_v = G_{v_L} \otimes G_{v_R}$. In fact, the proofs of Lemma 3.6, Lemmas 3.8–3.9, and Lemmas 3.11–3.14 are constructive proofs. By using these constructive proofs, our algorithm constructs a minimum and (n, s, t)-constrained path cover $\mathcal{PC}_v$ of G_v satisfying that $\mathcal{PC}_v$ contains at least one semi-terminal path except $\pi(G_v, \mathcal{T}_v) = \lceil \frac{|\mathcal{T}_v|}{2} \rceil$ or ($s_L = |\mathcal{T}_{v_R}| = 0$ and $|V_R| < n_L$). Lemma 3.10 shows that there exists no minimum constrained path cover of G_v with semi-terminal path if $s_L = |\mathcal{T}_{v_R}| = 0$ and $|V_R| < n_L$. Hence, $\mathcal{PC}_v$ is a minimum and (n, s, t)-constrained path cover of G_v satisfying that there exists no minimum constrained path cover of G_v with semi-terminal path if $s = 0$. Let $\widehat{E}_v = \{(u_L, u_R) | \forall u_L \in V_L$ and $\forall u_R \in V_R\}$. In the constructive proofs of Lemma 3.6, Lemmas 3.8–3.9, and Lemmas 3.11–3.14, we can easily see that $\mathcal{PC}_v$ can be constructed in $\mathrm{O}(|\widehat{E}_v|)$ time. Hence, we conclude the following theorem:

Theorem 3.16. *Given a cograph $G = (V, E)$ and a terminal set $\mathcal{T}$ of G, the terminal path cover problem can be solved in $\mathrm{O}(|V| + |E|)$-linear time.*

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